

How to succeed

- Read the Chapter ahead of lectures
- Come to class
- **Start paper early**
- Study groups
- Prepare exams
- Don't cheat
- **Learn with Internet: Wikipedia, google(baidu)...**
- Ask questions and Think skeptically

Last Week

- **Adhesives**

Introduction to Polymer Materials

Lecture 10: Fibers

Dazhi Sun

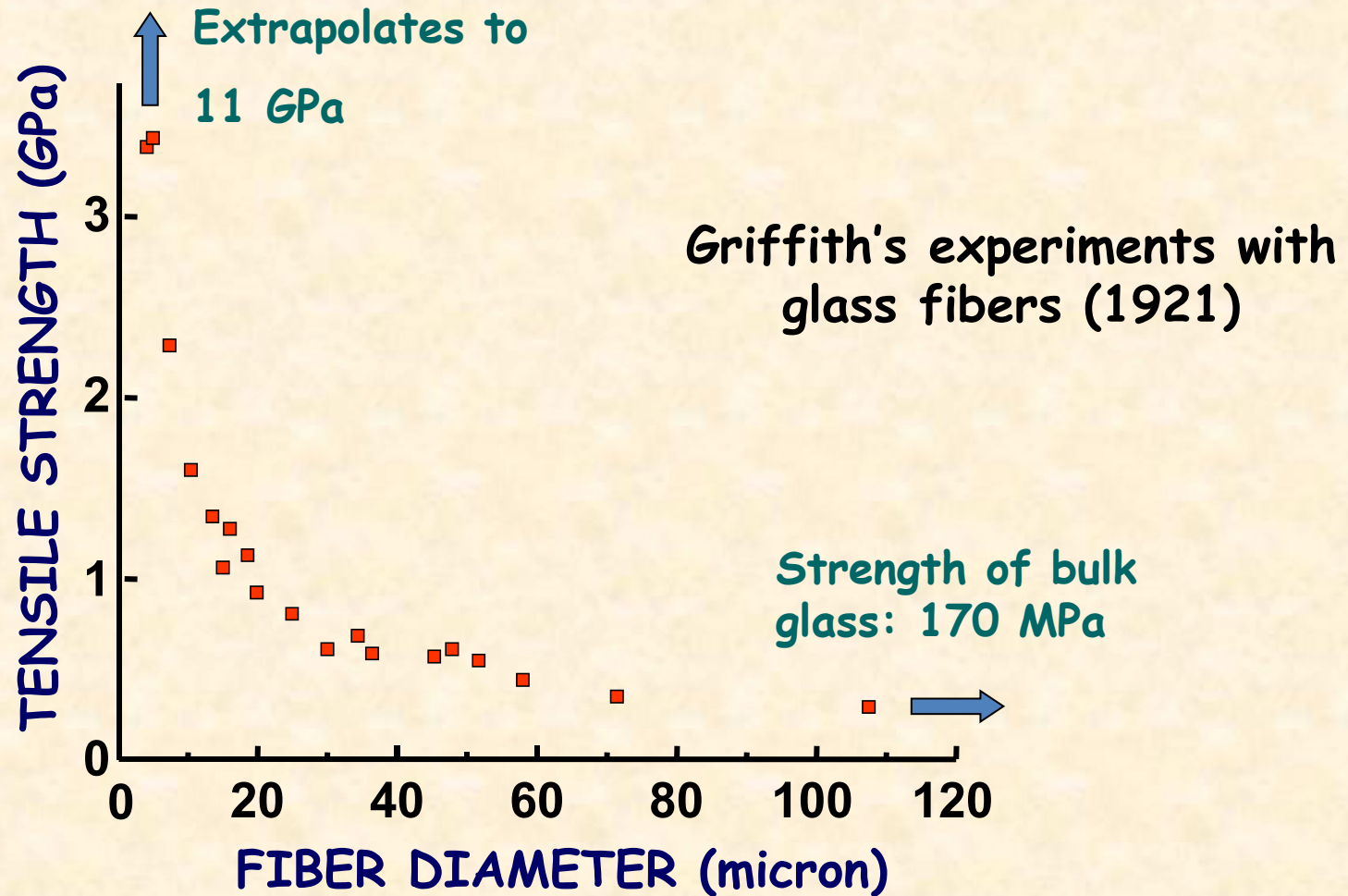
Polymer Fibers



Fibers

- A Fiber is a long, thin thing!
 - Aspect ratio >100
 - At diameters $> 75 \mu$, the fiber is a rod
- Long means:
 - > 1 kilometer
 - At a density of 1.4 and a denier of 5, 1 kilometer weighs less than 5 grams
 - > 1 kilogram
 - 1.5 kilograms at 5 dpf is 20,000 miles
- Few commercial fibers are produced at a scale of less than 500 tons
 - The length at 5 dpf is $\sim .01$ lightyear
- Typical melt spinning speeds are in excess of 100 miles/hour
 - To be viable, polymer to fiber conversions must be $\sim 90\%$

MACROSCALE vs MICROSCALE



Griffith's equation for the strength of materials

$$\sigma = \left(\frac{2E\gamma}{\pi a} \right)^{1/2}$$

a = length of defect

γ = surface energy

- Thus, going from the macroscale to the atomic scale (via the nanoscale), defects progressively become smaller and/or are eliminated, which is why the strength increases (see equation).
- Note that the Griffith model predicts that defects have no effect on the modulus, only on strength
- But note: the model also predicts that defects of zero length lead to infinitely strong materials, an obvious impossibility!

Polymer Fibers

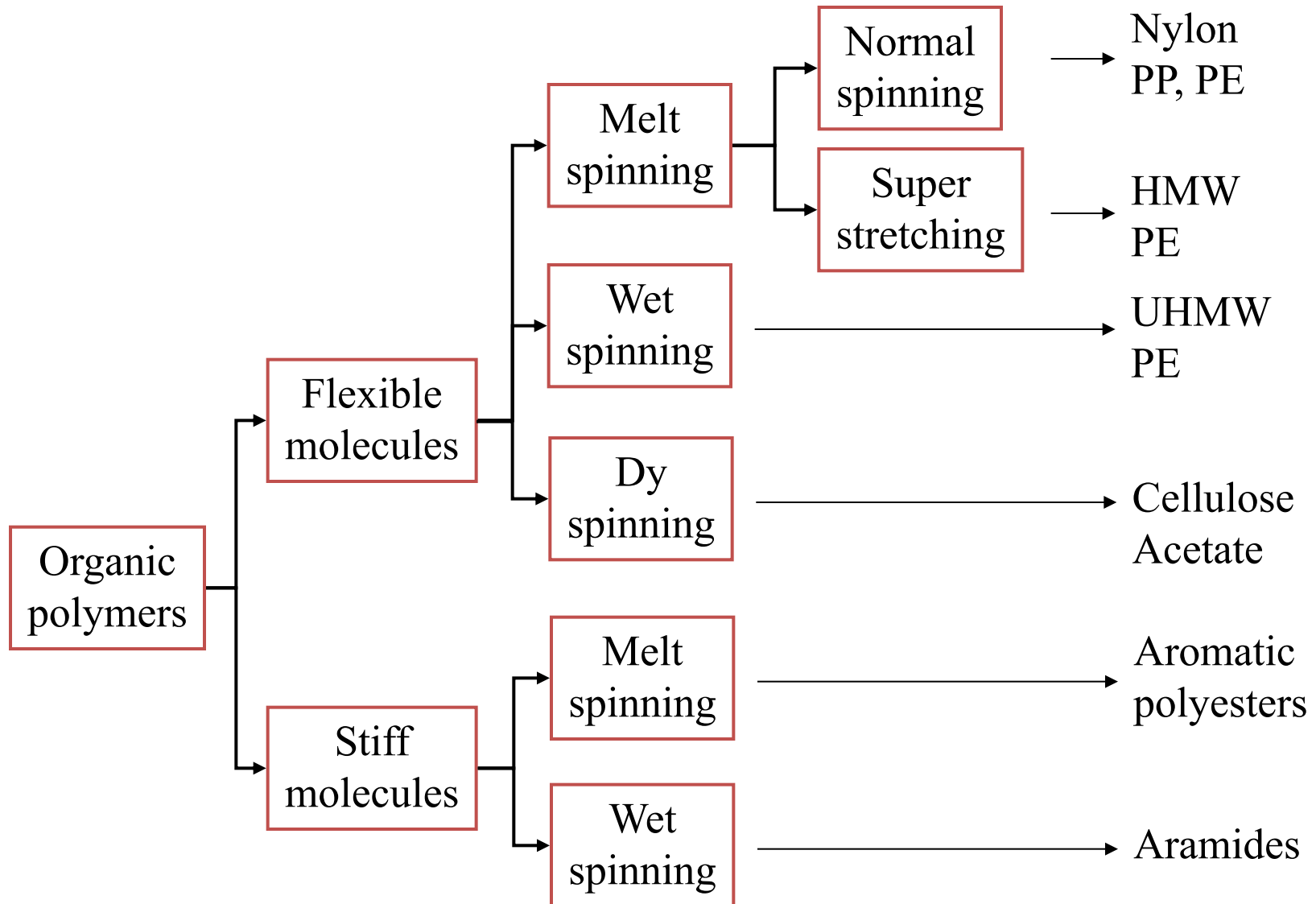
- **1000 X longer than diameter**
- **Kevlar-strongest organic fiber**

- Melt spinning technology can be applied to polyamide (Nylon), polyesters, polyurethanes and polyolefins such as PP and HDPE.
- The drawing and cooling processes determine the morphology and mechanical properties of the final fiber. For example ultra high molecular weight HDPE fibers with high degrees of orientation in the axial direction have extremely high stiffness !!
- Of major concern during fiber spinning are the instabilities that arise during drawing, such as brittle fracture and draw resonance. Draw resonance manifests itself as periodic fluctuations that result in diameter oscillation.

TABLE 4.2. Fiber Properties^a

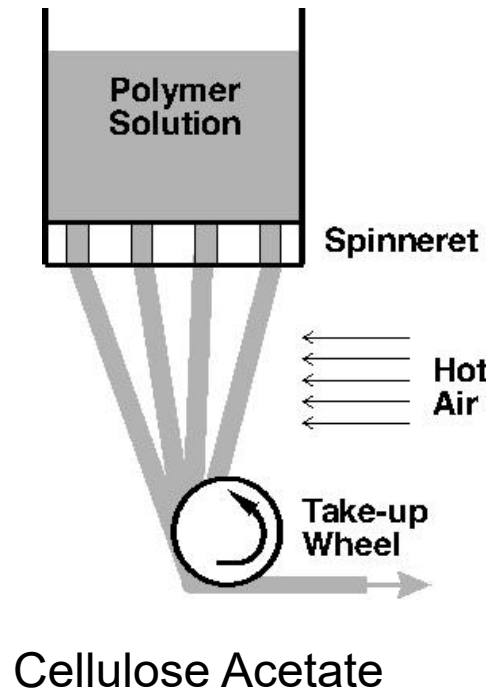
<i>Fiber Type</i>	<i>Tenacity^b</i> <i>(N/tex)</i>	<i>Specific</i> <i>Gravity</i>
Natural		
Cotton	0.26-0.44	1.50
Wool	0.09-0.15	1.30
Synthetic		
Polyester	0.35-0.53	1.38
Nylon	0.40-0.71	1.14
Aromatic polyamide (aramid) ^c	1.80-2.0	1.44
Polybenzimidazole	0.27	1.43
Polypropylene	0.44-0.79	0.90
Polyethylene (high strength)	2.65 ^d	0.95
Inorganic^c		
Glass	0.53-0.66	2.56
Steel	0.31	7.7

Polymer fibers – Fabrication

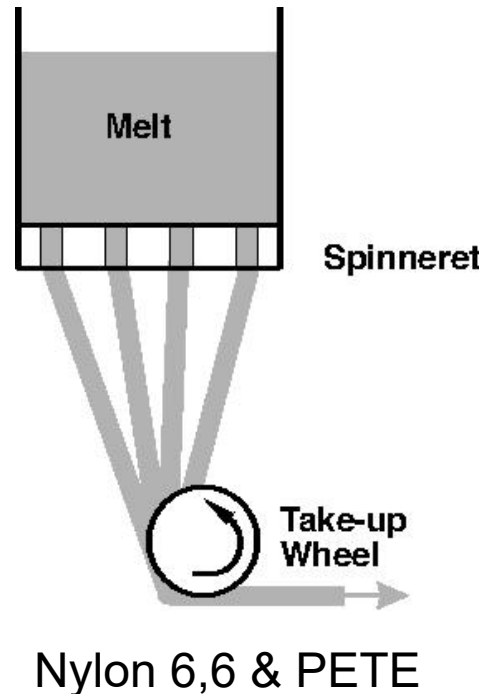


Fabrication

Dry Spinning:
From solution



Melt Spinning:
From Melt



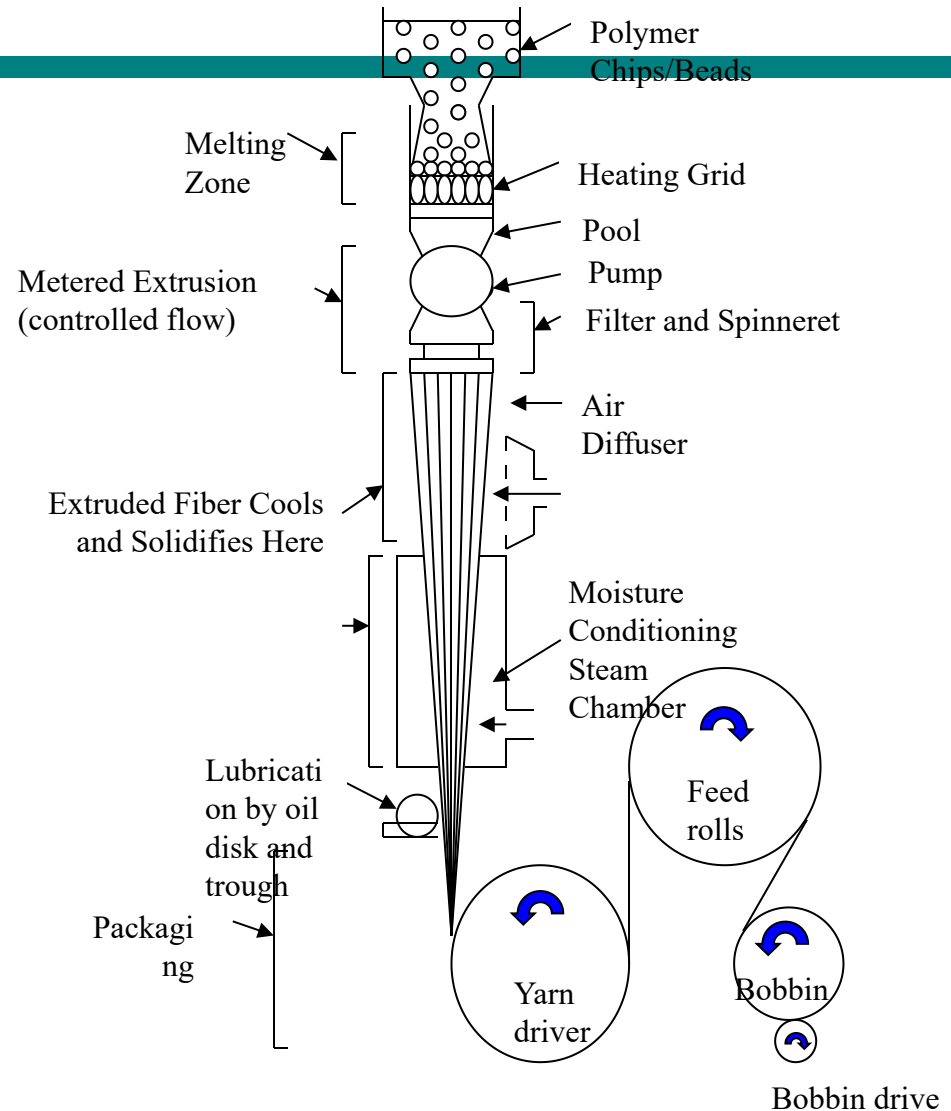
Wet Spinning:
From solution into
solution

Kevlar, rayon, acrylics,
Aramids, spandex



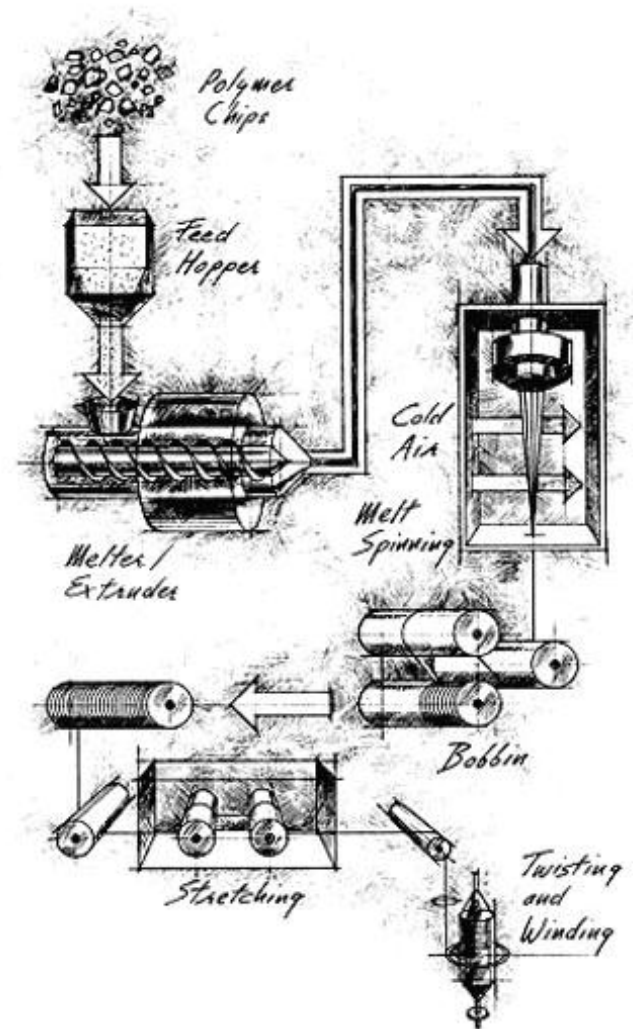
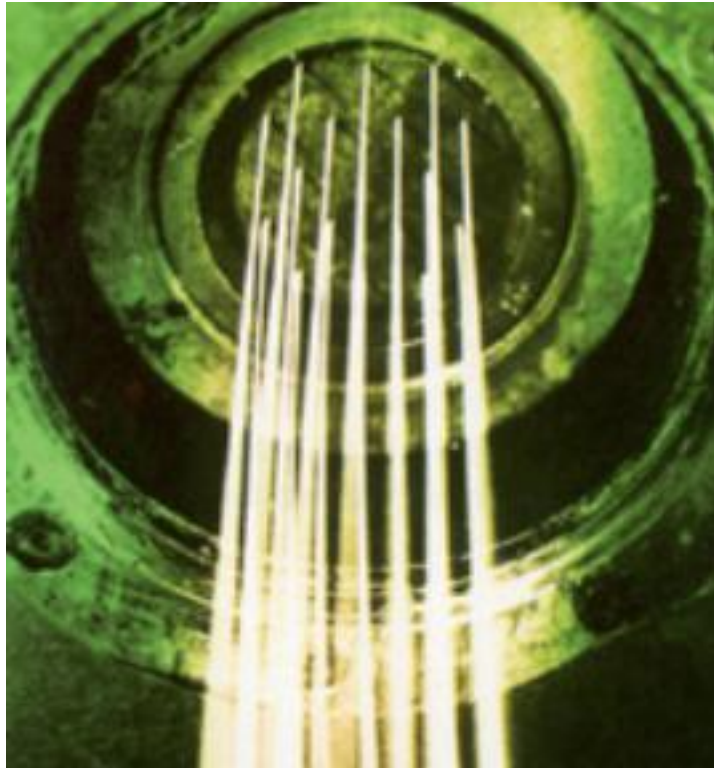
Fiber Spinning: Melt

Fiber spinning is used to manufacture synthetic fibers. A filament is continuously extruded through an orifice and stretched to diameters of 100 μm and smaller. The molten polymer is first extruded through a filter or “screen pack”, to eliminate small contaminants. It is then extruded through a “spinneret”, a die composed of multiple orifices (it can have 1-10,000 holes). The fibers are then drawn to their final diameter, solidified (in a water bath or by forced convection) and wound-up.



Nylon 6,6 & PETE

Melt spinning

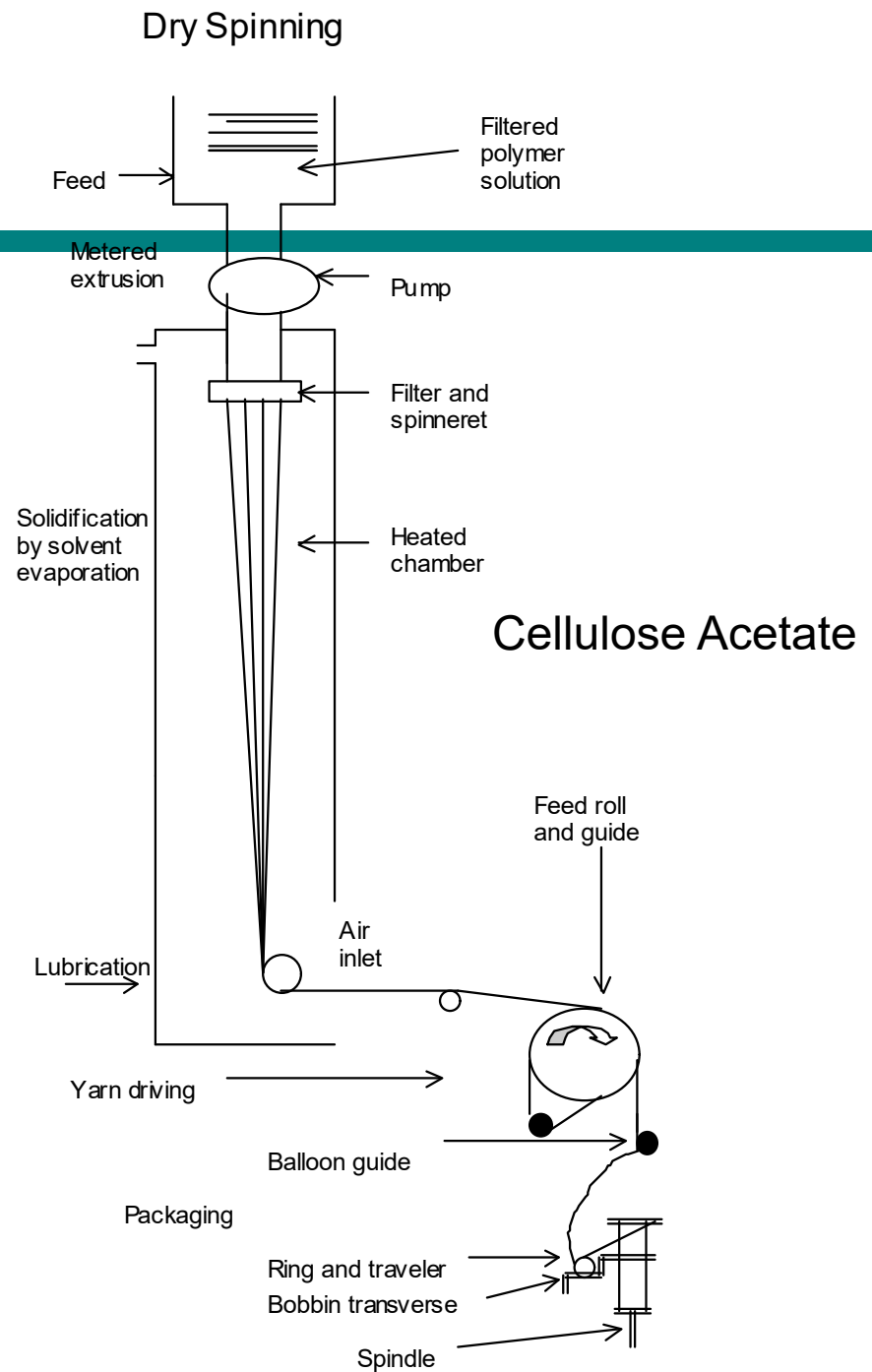


Melt Spinning Polymer from Chip

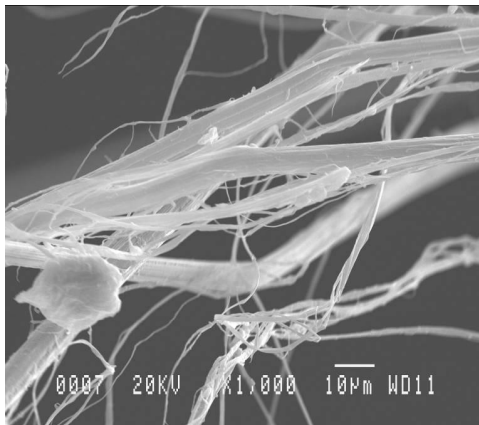
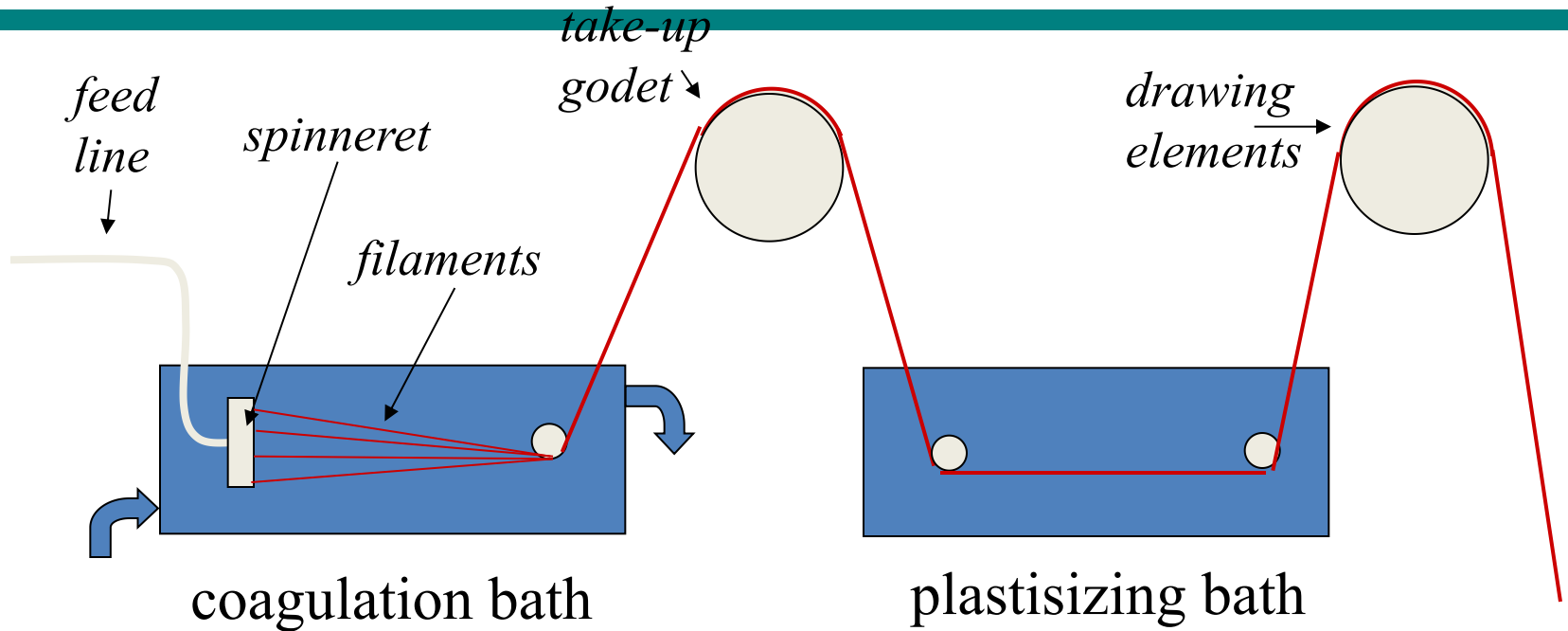
Dry Spinning of Fibers from Solution



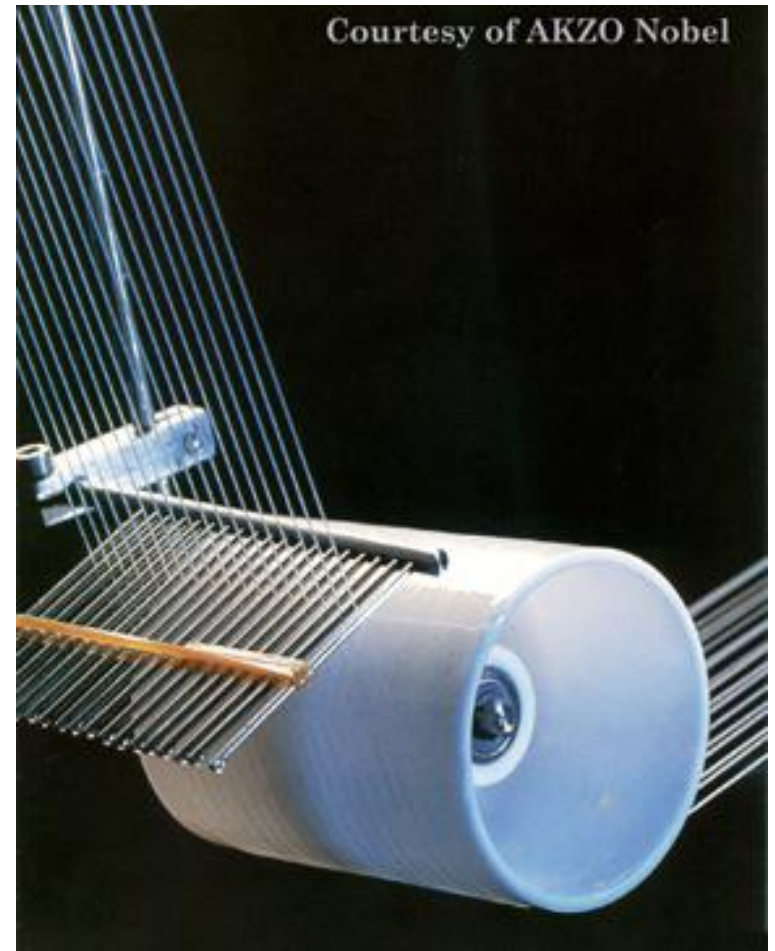
[Wiki](#)



Wet Spinning (e.g. Kevlar)

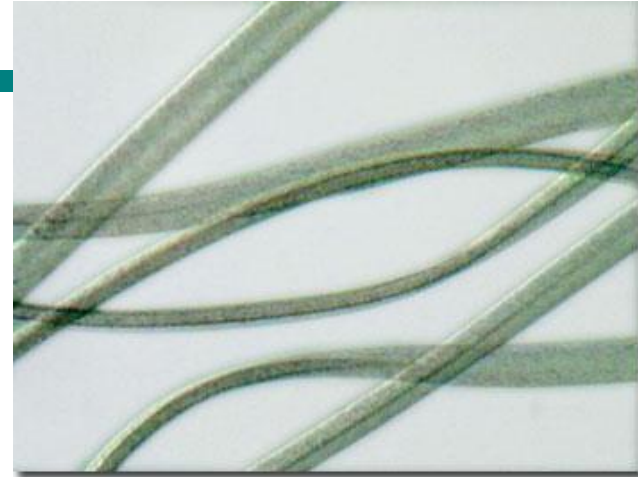


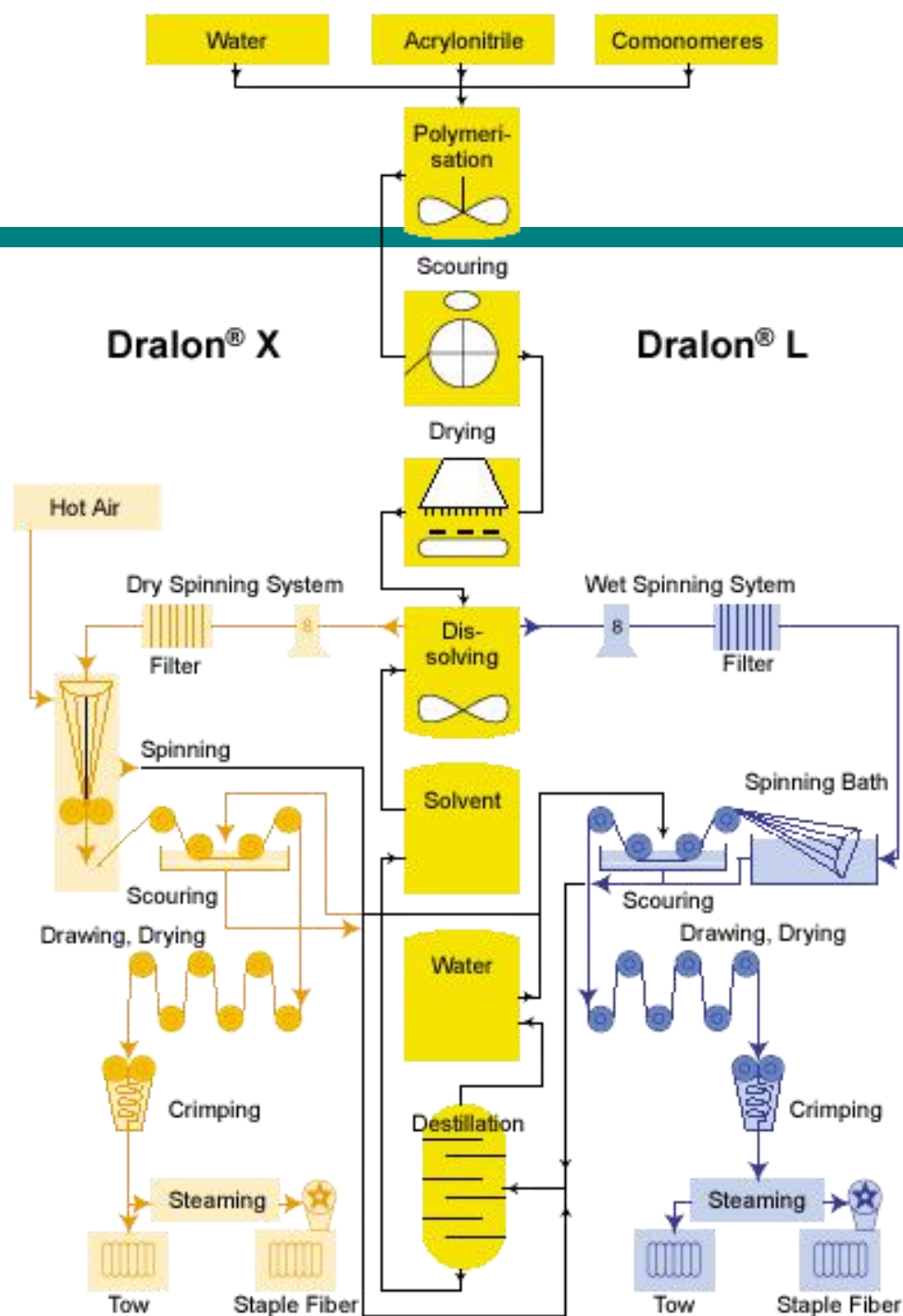
Kevlar, rayon, acrylics
Aramids, spandex



Acrylic Fibers

- 85% acrylonitrile
- Wet spun
- Acrylic's benefits are:
 - • Superior moisture management or wickability •
 - Quick drying time (75% faster than cotton) •
 - Easy care, shape retention •
 - Excellent light fastness, sun light resistance •
 - Takes color easily, bright vibrant colors •
 - Odor and mildew resistant

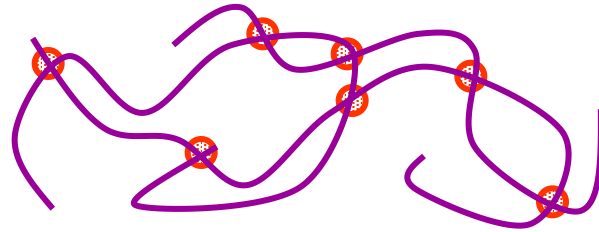




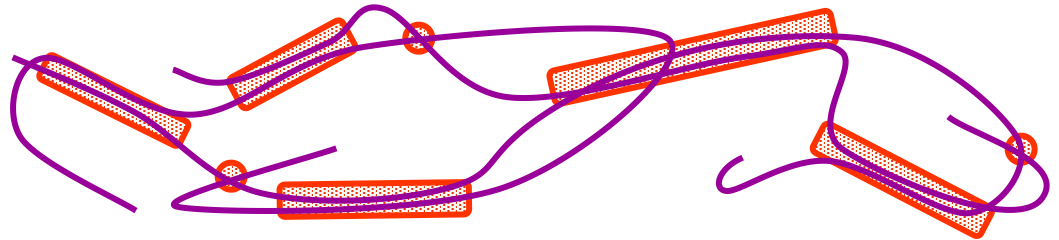
Why are fibers strong?

The source of strength: **van der Waals forces**

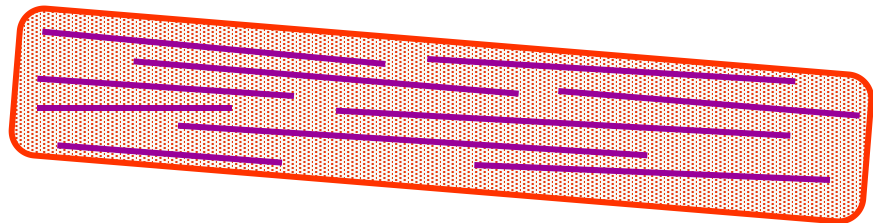
Flexible molecules,
normally spun



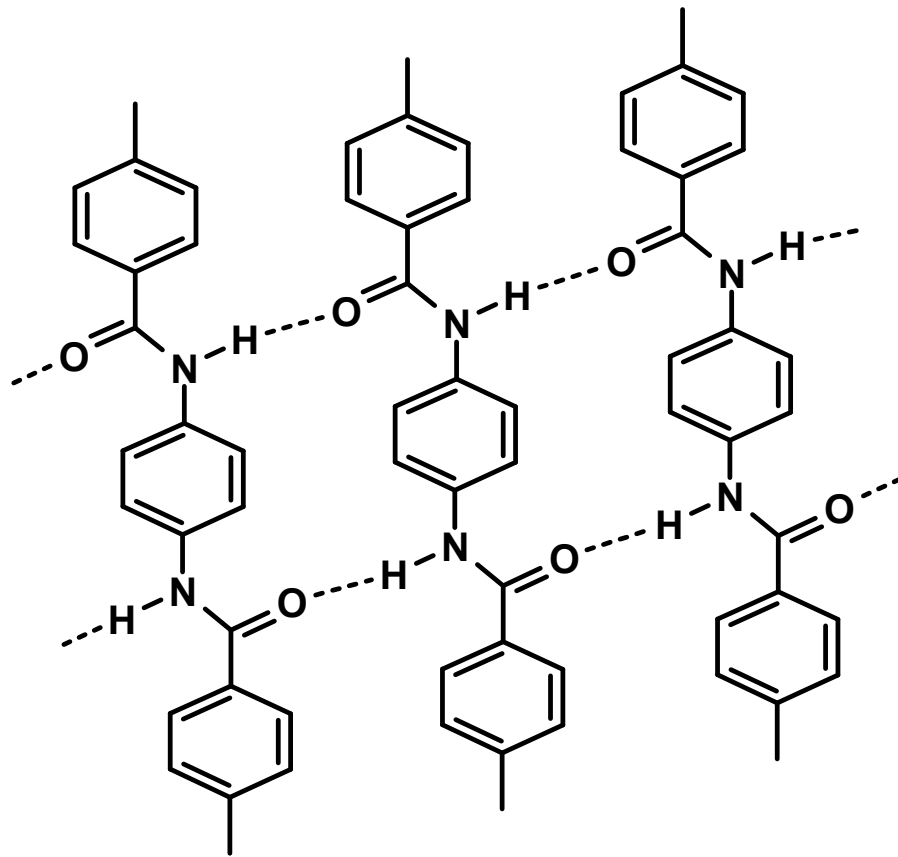
Flexible molecules
ultra stretched



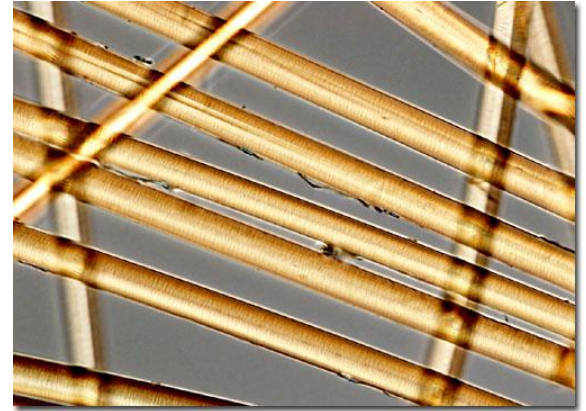
Rigid molecules
liquid crystallinity



Kevlar



Fiber orientation



Kevlar or Twaron



- High Tensile Strength at Low Weight
- Low Elongation to Break High Modulus (Structural Rigidity)
- Low Electrical Conductivity
- High Chemical Resistance
- Low Thermal Shrinkage
- High Toughness (Work-To-Break)
- Excellent Dimensional Stability
- High Cut Resistance
- Flame Resistant, Self-Extinguishing



Polypropylene elastomers

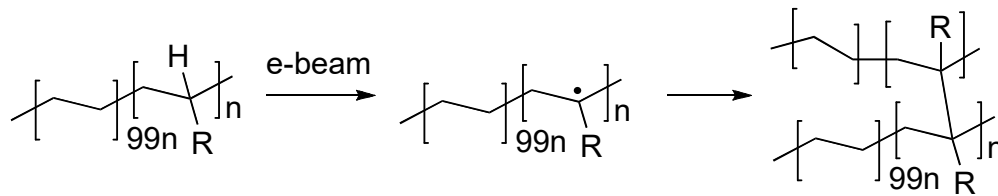
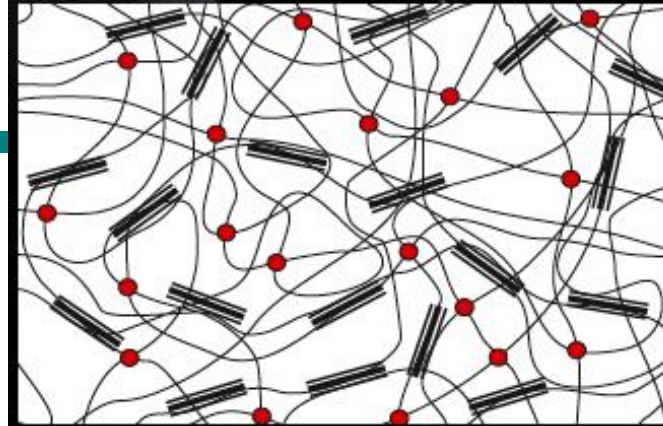


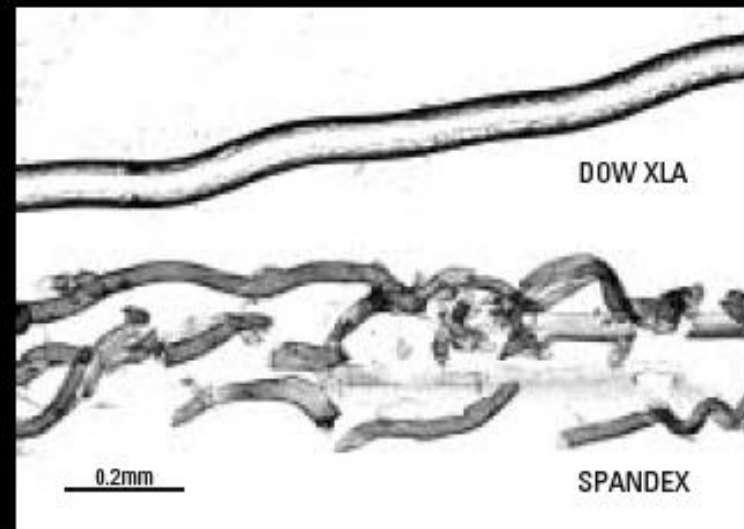
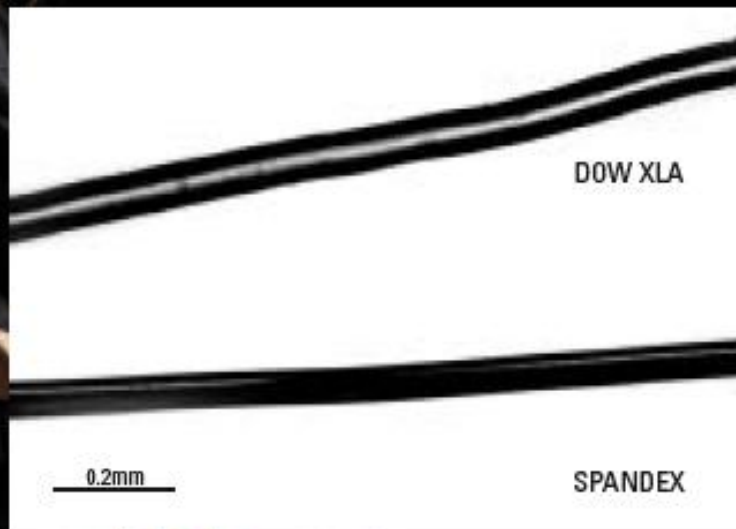
FIGURE 1: MICROSTRUCTURE OF DOW XLA FIBER



— Crystal

● Crosslink

FIGURE 5: HOT-STAGE PHOTOMICROGRAPHS OF 40-DENIER DOW XLA FIBER AND SPANDEX



DOW XLA Fiber Retains Thermal Stability Up to 220°C

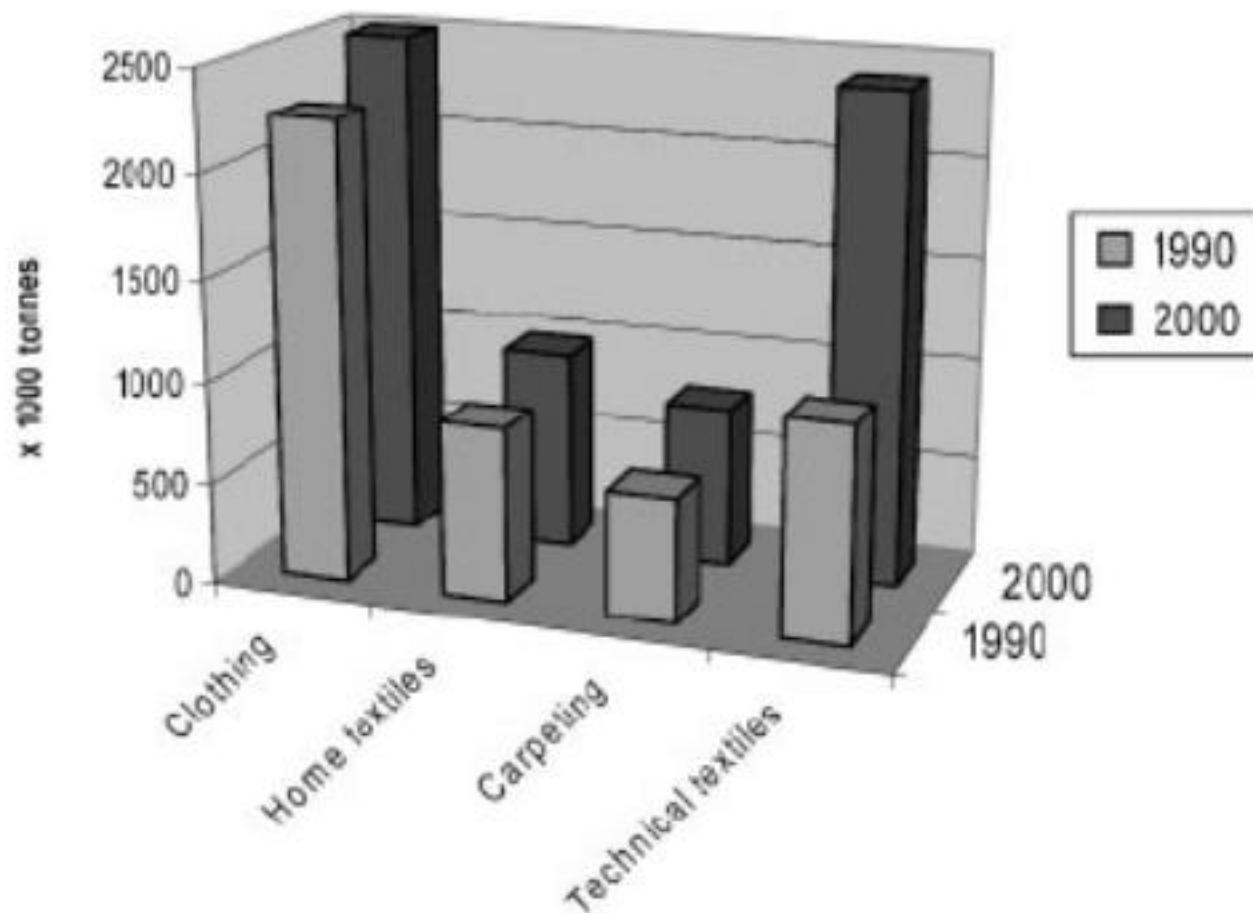
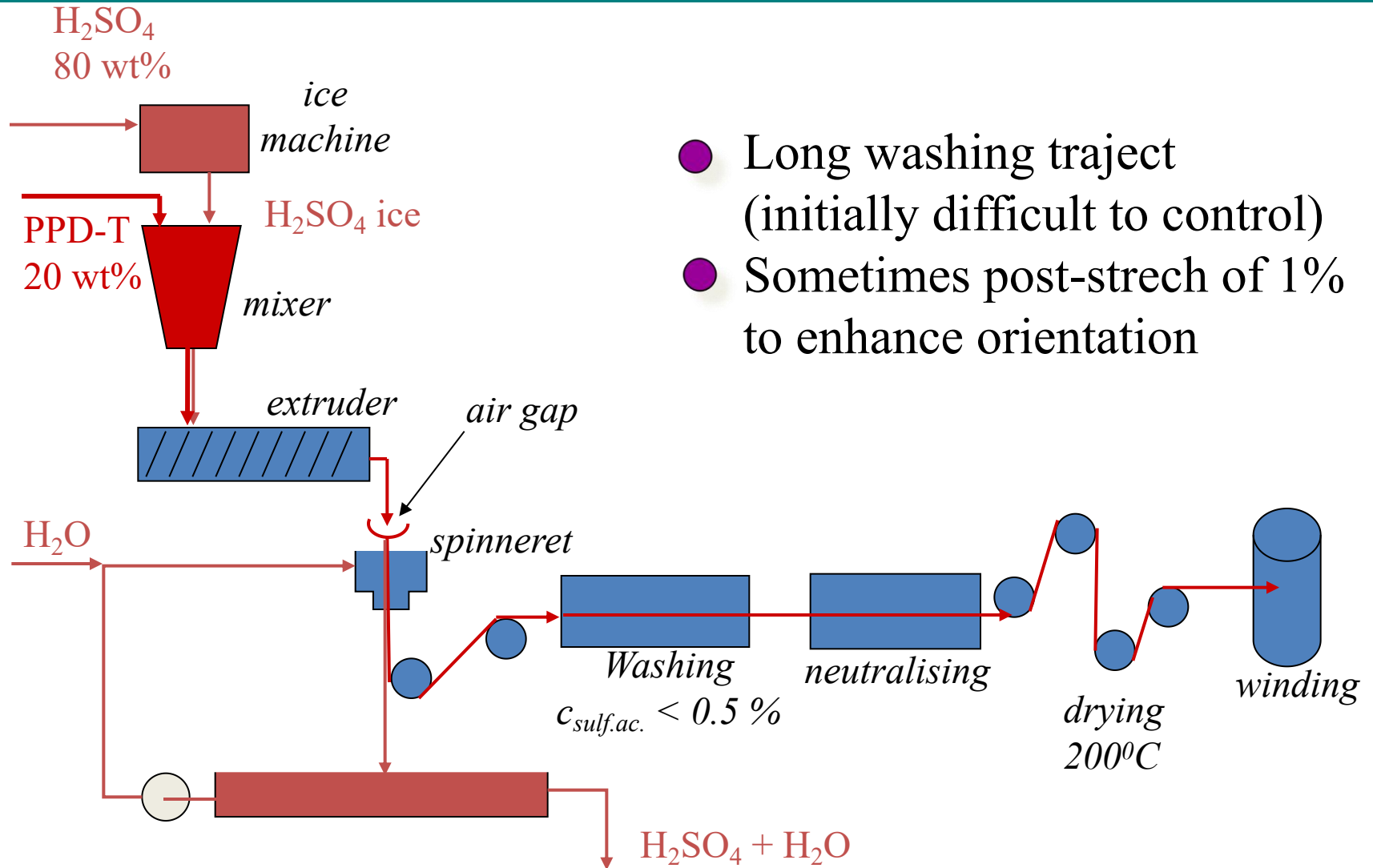


Figure 1. Consumption by the full European textile chain: evolution of technical textiles versus others sectors.

Aramide fibers

the complete spinning line



Strong fibers from flexible chains

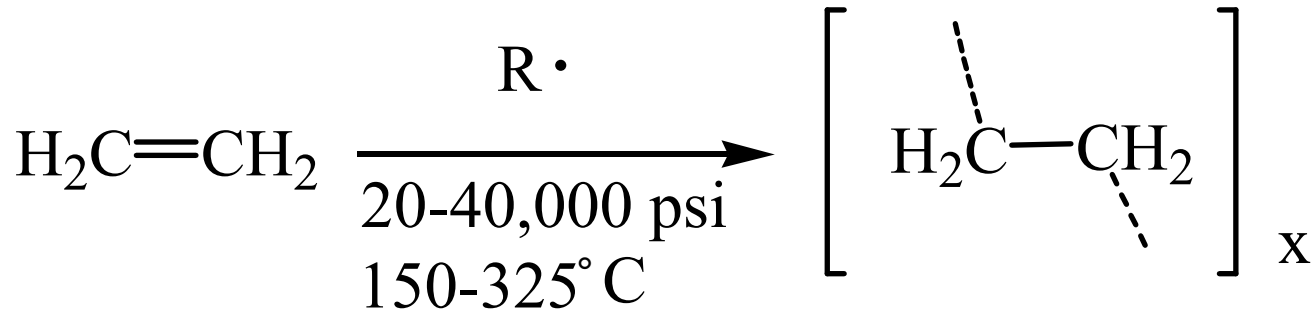
Super-stretched polyethylene:

- $M_w = 10^5$ (just spinnable)
- conventional melt spinning
- additional stretching of 30 to 50 times
- below the melting point

Wet (gel) spinning of polyethylene

- $M_w = 10^6$ (to high elasticity for melt spinning)
- decalin or parafin as solvent
- formation of thick (weak) fibers without stretching
- removal of the solvent
- stretching of 50 to 100 times close to melting point

Polyethylene (LDPE)

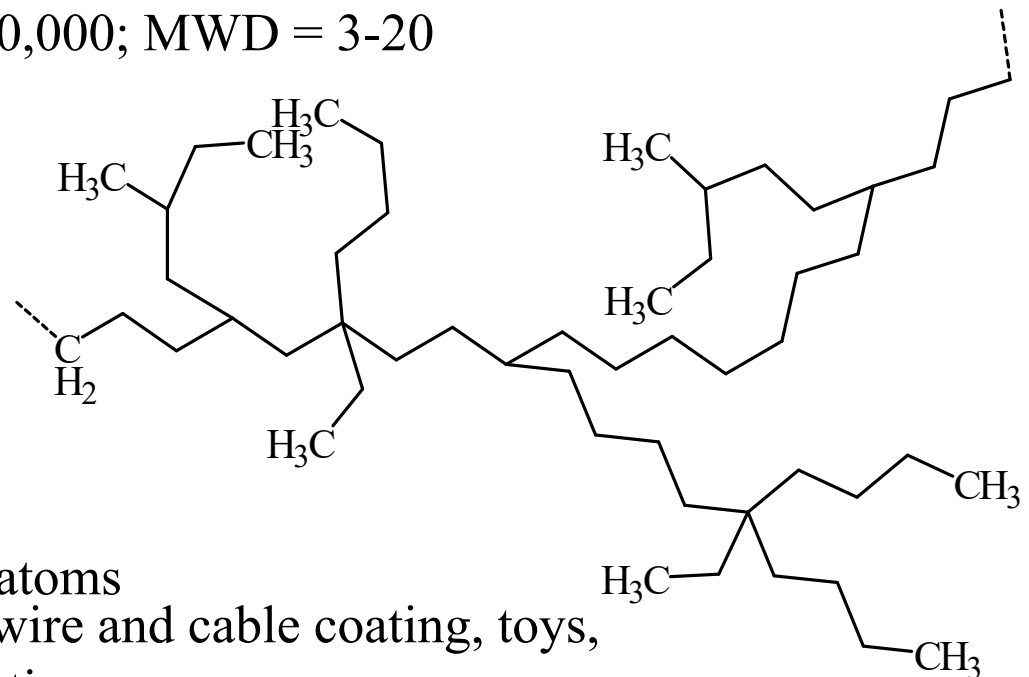


Molecular Weights: 20,000-100,000; MWD = 3-20
density = 0.91-0.93 g/cm³

Highly branched
structure—both long and
short chain branches

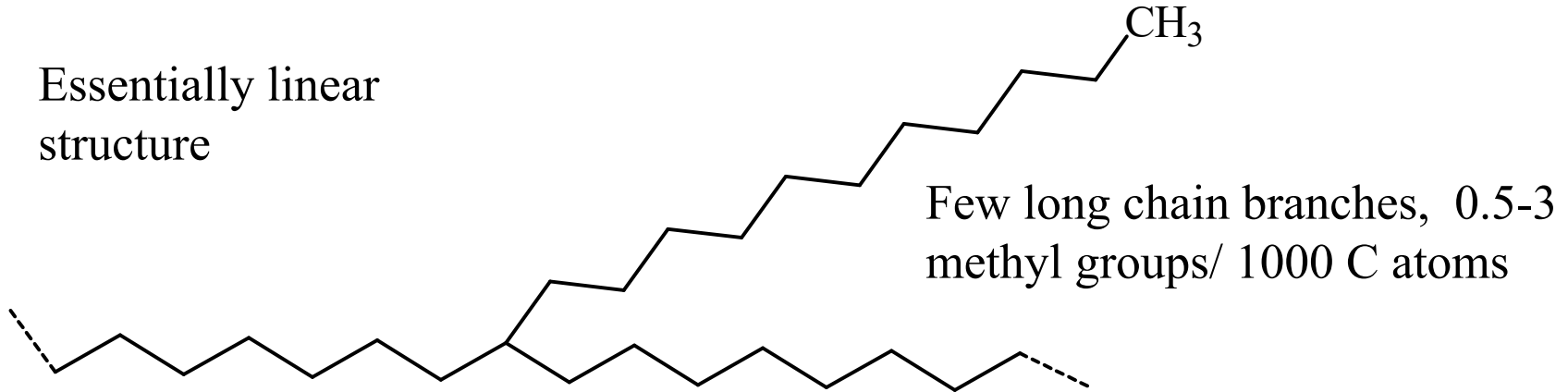
T_m ~ 105 C, X'linity ~ 40%

15-30 Methyl groups/1000 C atoms
Applications: Packaging Film, wire and cable coating, toys,
flexible bottles, housewares, coatings



Polyethylene (HDPE)

Essentially linear
structure



Molecular Weights: 50,000-250,000 for molding compounds

250,000-1,500,000 for pipe compounds

>1,500,000 super abrasion resistance—medical implants

MWD = 3-20

density = 0.94-0.96 g/cm³

T_m ~ 133-138 C, X'linity ~ 80% Generally opaque

Applications: Bottles, drums, pipe, conduit, sheet, film

UHMWPE fibers: Dyneema



Gel spinning process

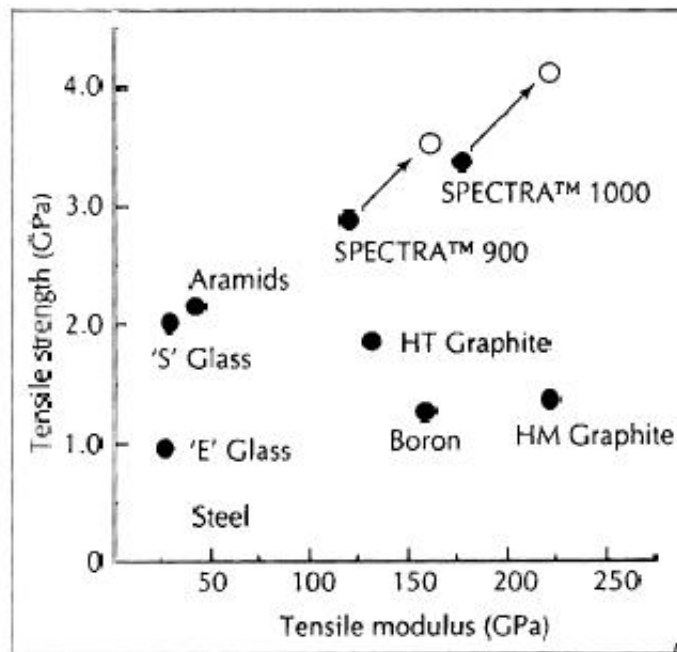
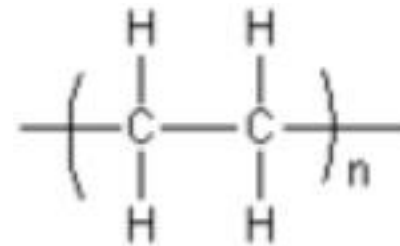


Fig. 3 The specific strength and modulus of various reinforcing fibers. The unique effect of deformation rate on these properties for USPE fibers is shown: ●, low rate (10^{-1} min^{-1}); ○, high rate (10^1 min^{-1}) (reproduced, with permission, from Ref. 33).



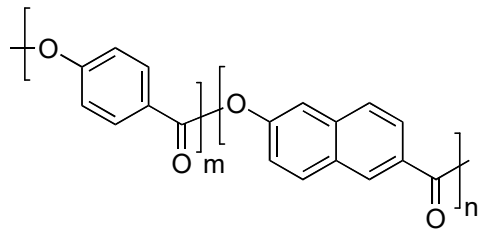
Structure of UHMWPE,
with $n = 100,000\text{-}250,000$

<http://www.dyneema.com>

Comparison of mechanical properties

	Strength (Gpa)	Modulus (Gpa)	stretch (%)
Classical fibres			
• nylon	1.0	5.6	18
• glass	2.7	69	2.5
• steel	2.8	200	2
Strong fibres			
• superstretched PE	0.7	4.7	
• wet spun PE (Dyneema)	2.2	80	3.4
• melt spun PE (Vectran)	3.2	90	3.5
• wet spun aramide	2.7	72	3.3
• idem with post-stretch	3.6	130	2.3

Vectran



Vectran Fiber Products*

HS Products

Denier	DPF	Filament Diameter, Microns	Filament Counts	Yield, Yards/lb
2,250	5.0	23	450	2,000
1,500	5.0	23	300	3,000
1,000	5.0	23	200	4,460
750	5.0	23	150	6,000
400	5.0	23	80	11,150
200	5.0	23	40	22,300
100	5.0	23	20	44,600
25	5.0	23	5	178,400

M Products

Denier	DPF	Filament Diameter, Microns	Filament Counts	Yield, Yards/lb
400	5.0	23	80	11,150
900	4.5	21	200	4,970
900	5.0	23	180	4,970

* Cut fiber and pulp also available.



Vectran fiber is thermotropic, it is melt-spun, and it flows at a high temperature under pressure

Figure 1: Schematic Of Molecular Chain Structure Of Fiber

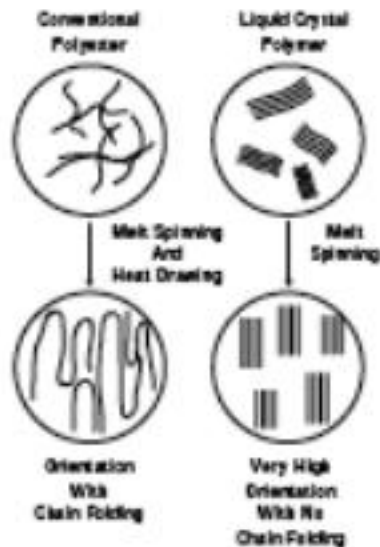


Table 1:

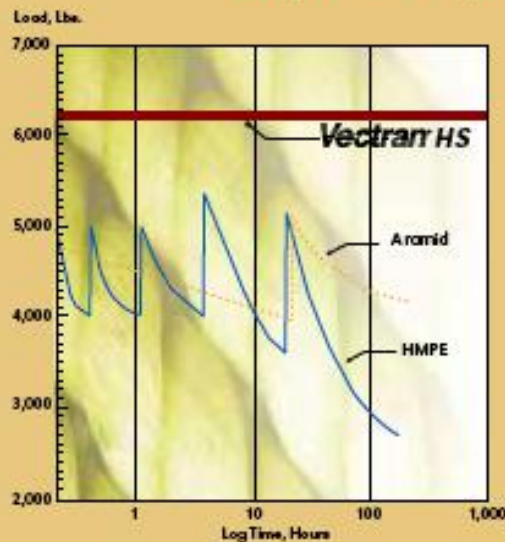
Comparison of Properties of Various Engineering Materials

Material	Density (g/cm ³)	Tensile Strength (GPa)	Specific Strength (km ^{**})	Tensile Modulus (GPa)	Specific Modulus (km ^{**})
Vectran®	1.4	2.9	210	65	4740
Titanium	4.5	1.3	29	110	2500
Stainless Steel	7.9	2.0	26	210	2700
Aluminum	2.8	0.6	22	70	2600
E-Glass	2.6	3.4	130	72	2800
Graphite (AS4)	1.8	4.3	240	230	13000

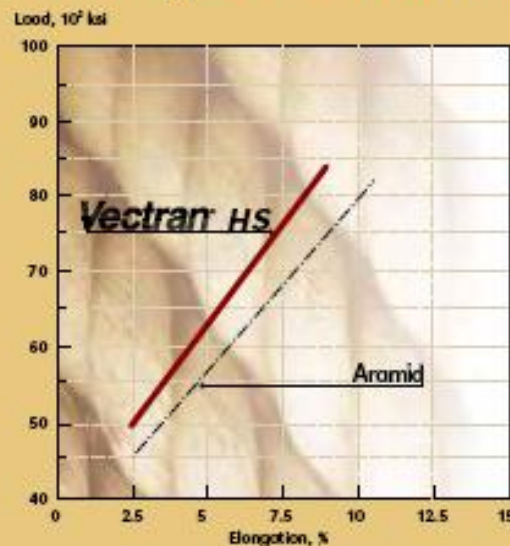
*Specific strength = Strength/Density (also divided by force of gravity for SI units). Also known as breaking length, the length of fiber that could be held in a vertical direction without breaking.

** Specific modulus = Modulus/Density (also divided by force of gravity for SI units). This measure increases with increasing stiffness and decreasing density.

Vectran HS Wire Rope Creep



Break Strength vs D/d Wire Rope Construction



Vectran HS Fiber Abrasion Comparison

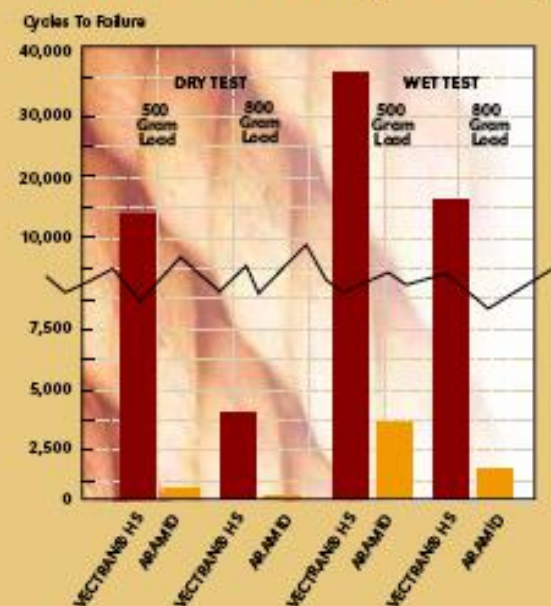
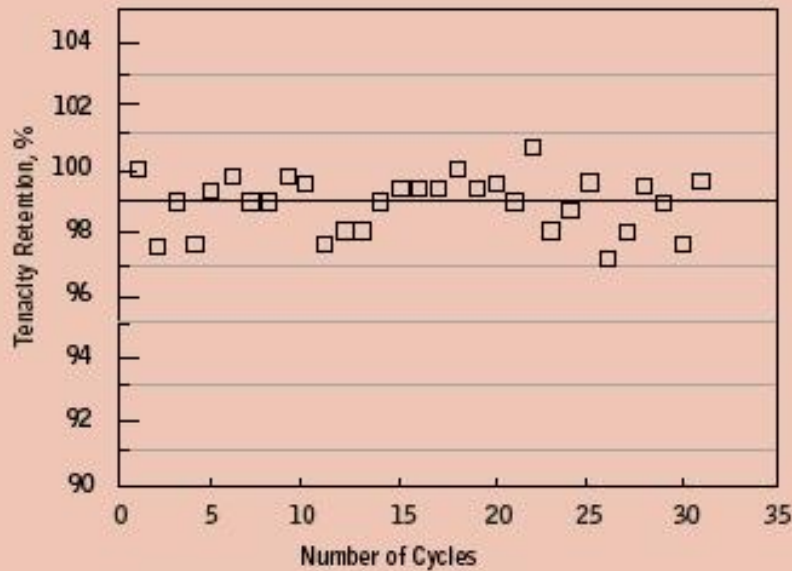
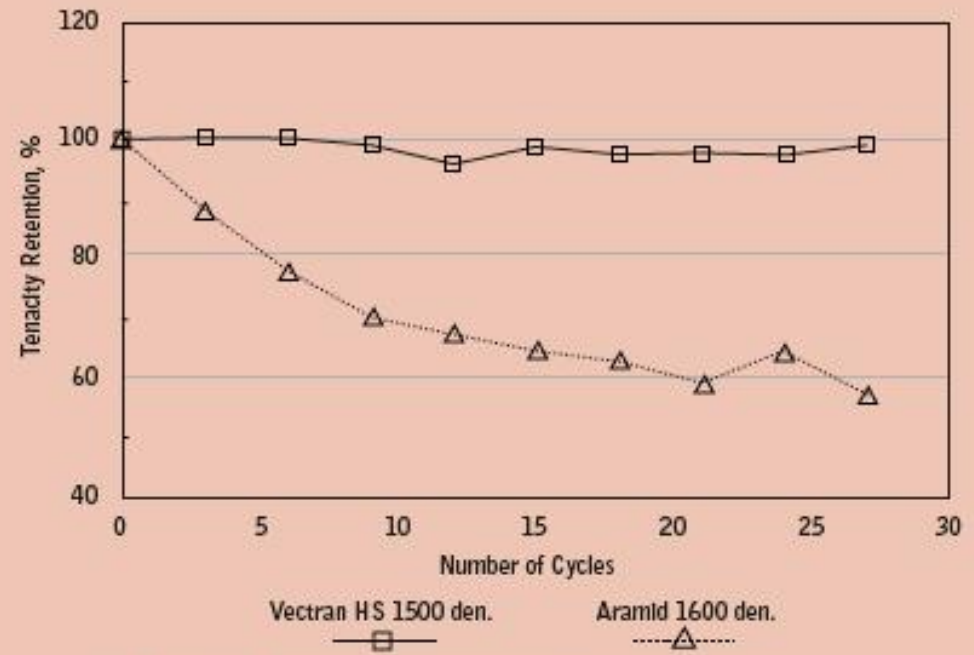


Figure 5: Vectran HS Tenacity vs Cycles At Temperature



Each Cycle = 8 hours @ 120°C followed by testing at ambient temperature

Figure 6: Vectran HS 1500/300 Filament Yarn Tenacity – Cycles at 195°C



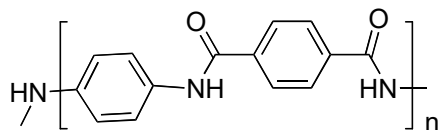
Each Cycle = 8 hours @ 195°C; testing at ambient temperature

Table 9:

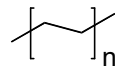
Comparative Testing of Yarn-on-Yarn Abrasion Resistance

Yarn	Average Cycles-to-Failure	
	Dry	Wet
Vectran T97, 1500D	16672	21924
Aramid 1, 1500D	1178	705
Aramid 2, 1500D	1773	759
Aramid 3, 1500D	974	486
PBO, 1500D	2153	—
HMPE, 1600D	8518	23619

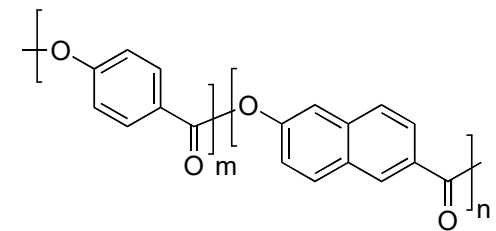
Test Method CI-1503: 1.5 wraps, 500g load, 66 cycles/min, no twist



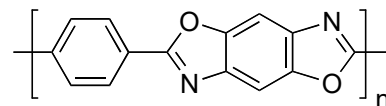
Aramid



Ultra High Molecular Weight Polyethylene



Vectran



poly(p-phenylene benzobisoxazole)

Zylon

Table 2. Description and properties of high temperature resistant fibres

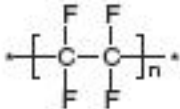
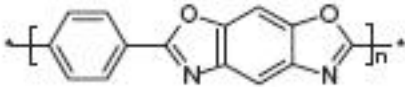
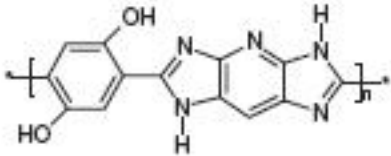
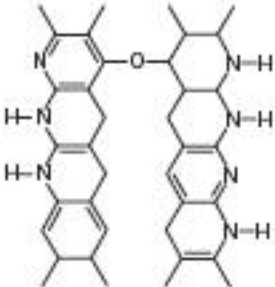
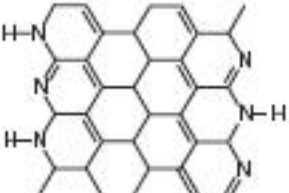
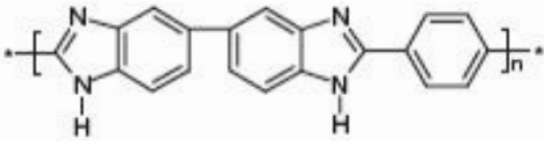
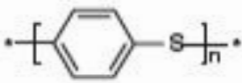
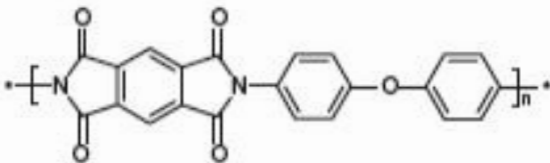
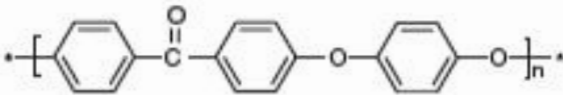
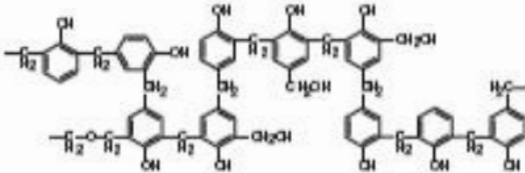
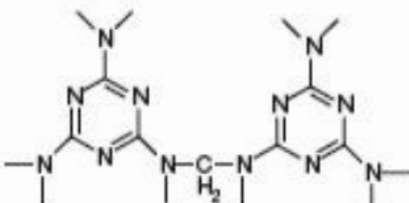
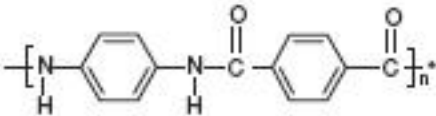
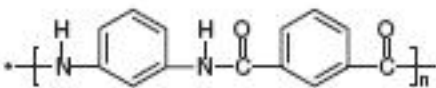
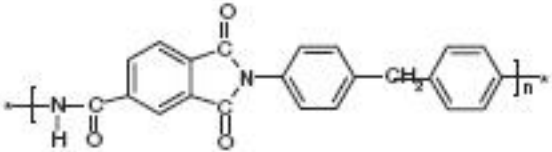
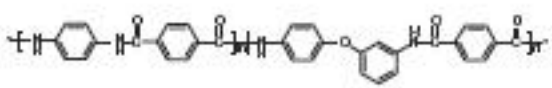
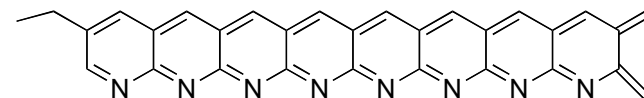
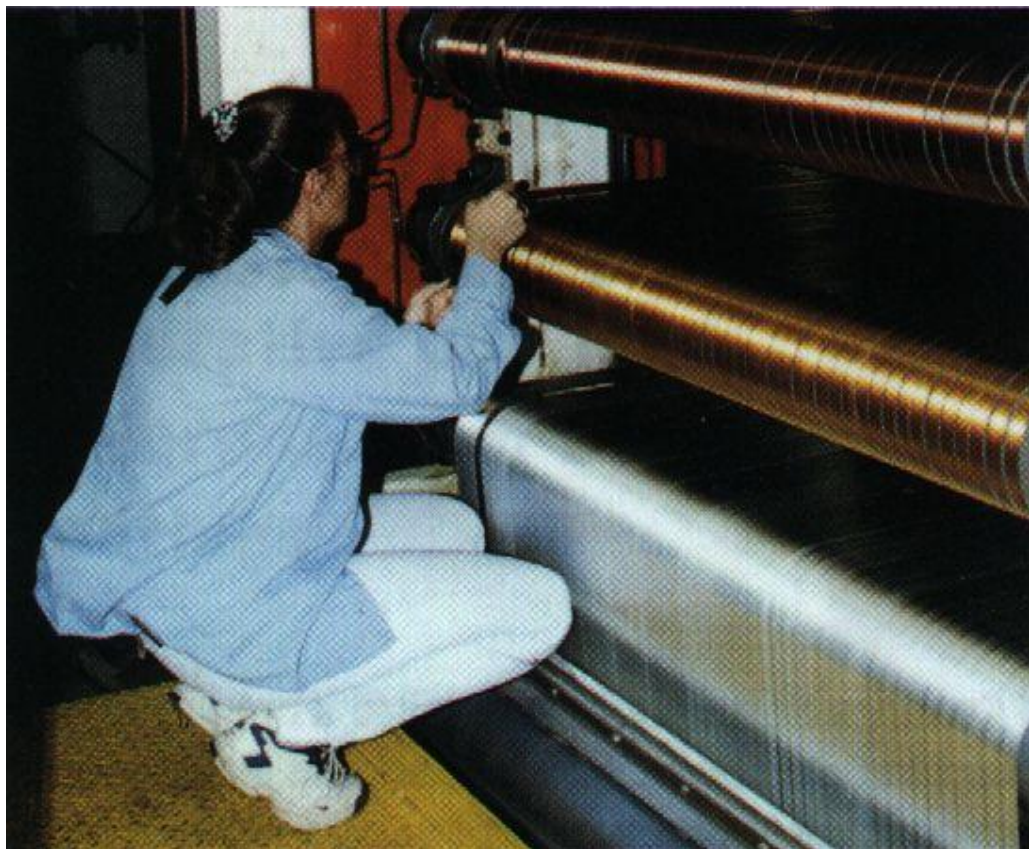
Chemical name or designation	Generic name	Chemical structure	Trade name	LOI (vol.-%)	Continuous operating temperature (°C)	Applications
Poly(tetra-fluoroethylene)	PTFE		Teflon [®] (DuPont) Toyoflon [®] (Toyobo)	95	260	Aerospace and automotive applications, food processing, agricultural machinery
Poly(p-phenylene-2,6-benzobisoxazole)	PBO		Zylon [®] (Toyobo)	56–68	310	Protective clothing, heat resistant felt, sling, fragmentation barrier
Poly(2,6-diimidazo(4,5-b:4',5'-e)pyridi-nylene-1,4 (2,5-dihydroxy)-phenylene)	PIPD		M5 (Magellan Systems International)	> 50	—	Protective clothing, composite materials, stealth materials, aerospace applications
Oxidized poly(acrylonitrile) fibre	OPF		Pyromex [®] (Toho Tenax), Lastan [®] (Asahi), PANOX [®] (SGL Carbon Group)	45–60	200	Aerospace structures, automotive engine, ballistic material, heating elements, thermal protection
Elongatable carbonaceous Dow fibre	EDF		EDF (Dow)	45–55	200	Insulative material

Table 2. (Continued)

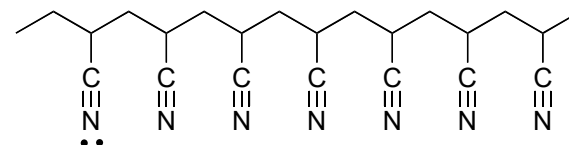
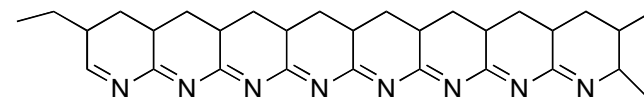
Poly(2,2'-(m-phenylene)-5,5'-bisbenzimidazole)	PBI		PBI (Celanese)	41	250	Firefighter protective clothing, industrial workers apparel, fire blockers
Poly(phenylene sulphide)	PPS		Ryton® (Phillips Petroleum Co.) Procon® (Toyobo)	~40	280	Reinforcing materials, filtration, protective clothing, insulators
Polyimide	PI		P84™ (Inspec Fibres)	38	260	Fire protective clothing, filtration
Poly(ether-etherketone)	PEEK		PEEK® (Zyex)	35	250	High temperature materials
Phenolic fibre	Phenolic or novoloid fibre		Kynol® (Nippon Kynol Inc)	30-34	200	Fire protective apparel, fire blockers, composite reinforcement
Melamine fibre	Melamine fibre	Three-dimensional melamine-based network 	Basofil® (BASF)	30-32	200	Fire protective clothing, insulative materials, fire blockers

Poly(p-phenylene terephthal-amide)	PPTA		Kevlar® (DuPont) Twaron (Teijin)	28–30	190	Protective clothing, composite reinforcement, ropes and cables
Poly(m-phenylene isophthal-amide)	PMIA		Nomex® (DuPont) Conex® (Teijin)	30–32	200	Fire protective clothing, filtration, thermal resistant furnishing
Polyamide-imide	PAI		Kermel® (Rhodia)	30–32	200	Fire and heat protective clothing, insulators
Copoly(p-phenylene-3,4-oxidi phenylene- terephthal-amide)	Copolymer p-aramid fibre		Technora® (Teijin)	25	200	Reinforced plastics, rope and cable, protective clothing, reinforcement of construction materials

Carbon Fibers: Pyrolyzing Polyacrylonitrile Fibers



Young's Modulus 325 Gpa
Tensile Strength 3-6 GPa



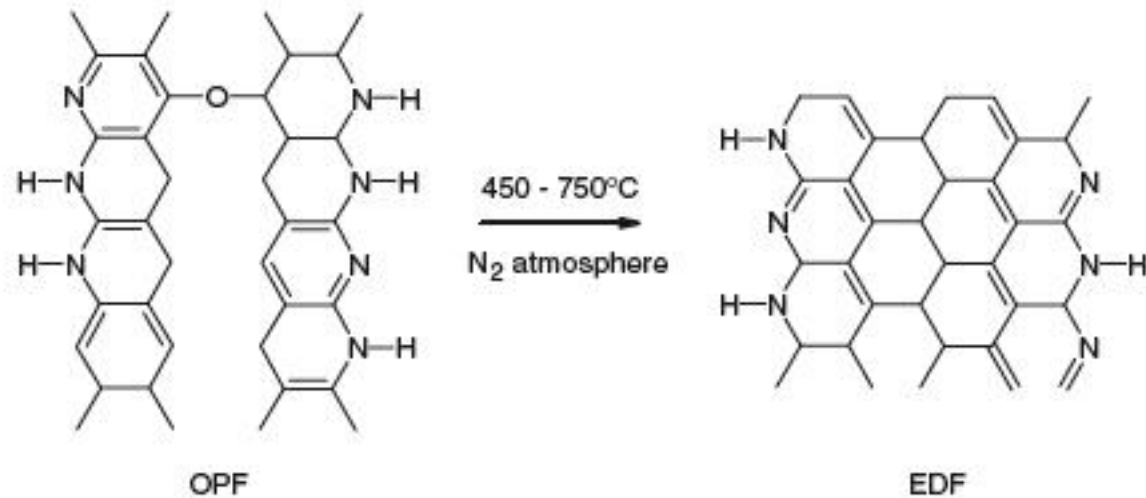
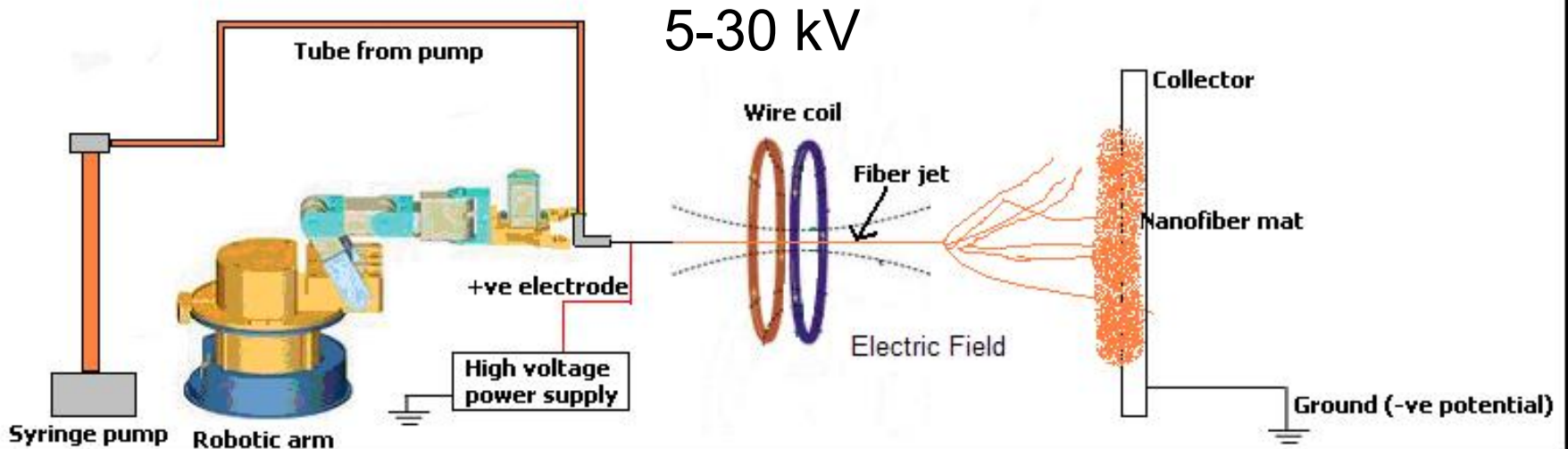


Figure 4. Synthesis of elongatable Dow fibre (EDF) from oxidized PAN fibre (OPF).

Electrospinning of Fibers



- Driving force is charge dissipation, opposed by surface tension
- Forces are low
- Level of charge density is limited by breakdown voltage – Taylor cone formation

Fiber diameter $\propto$ [Voltage]⁻¹

- “Inexpensive” and easy to form nanofibers from a solution of practically any polymer (Formhals 1934)
- Only small amount of material required

Increasing Electric Potential →

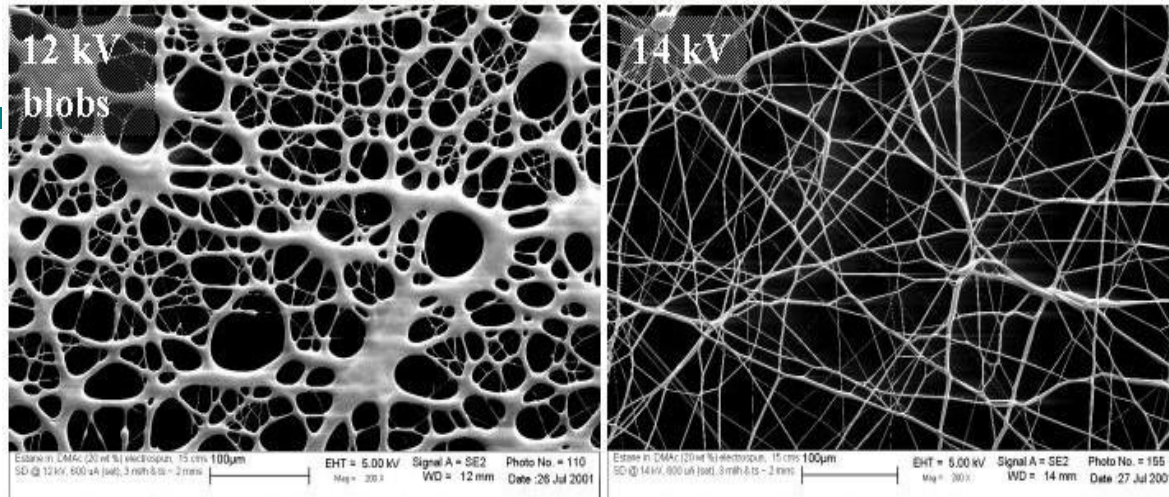


Figure 5. Effect of increasing electric potential on 20-wt % Estane® 5750 electrospun at 3ml/h & 15 cm capillary-screen distance. Fiber diameter decreases with increasing voltage

Increasing capillary-screen distance →

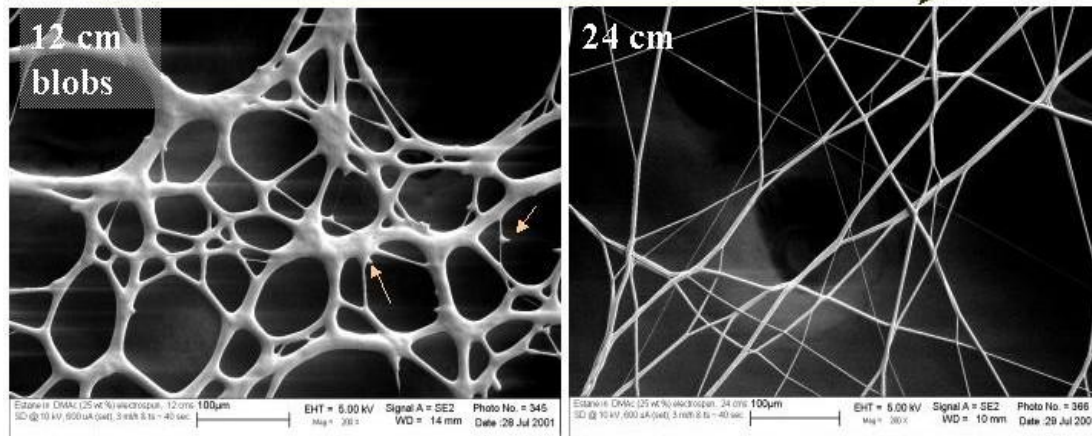


Figure 4. Effect of increasing capillary-screen distance on 25-wt % Estane® 5750 electrospun at 10 kV & 3ml/h. The Average diameter range is 5 μm – 905 nm. A broad distribution of fiber diameters was obtained.

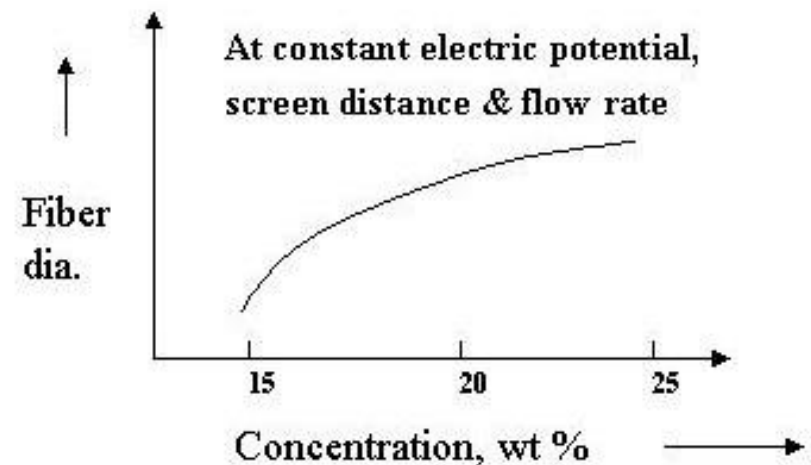
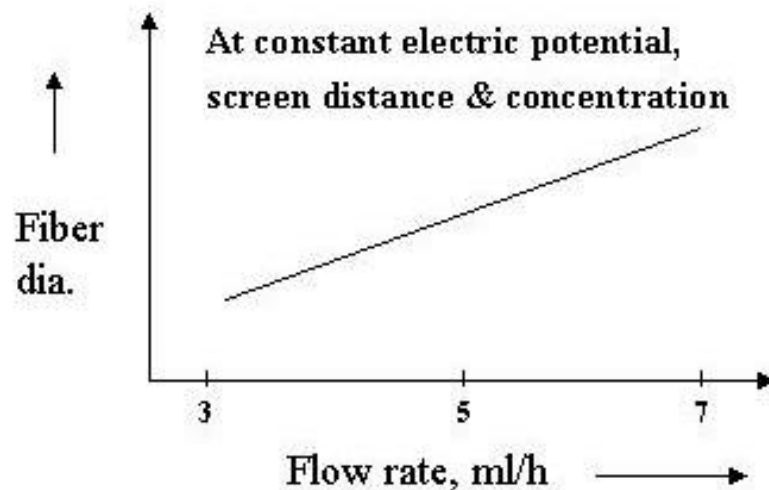
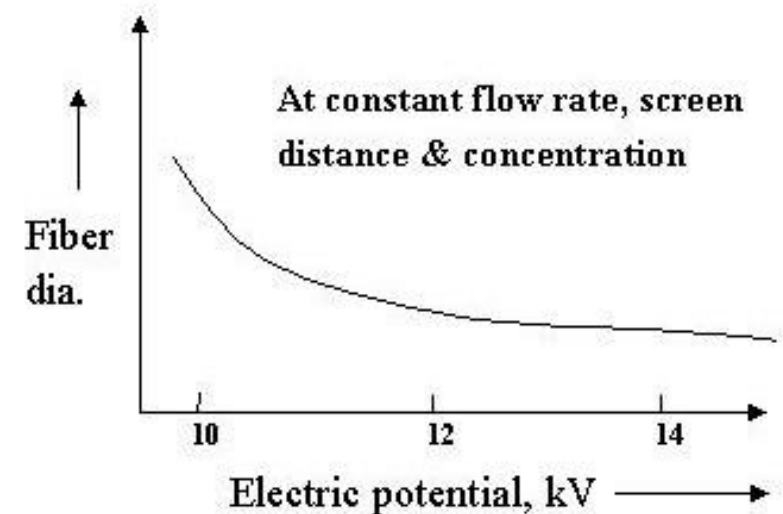
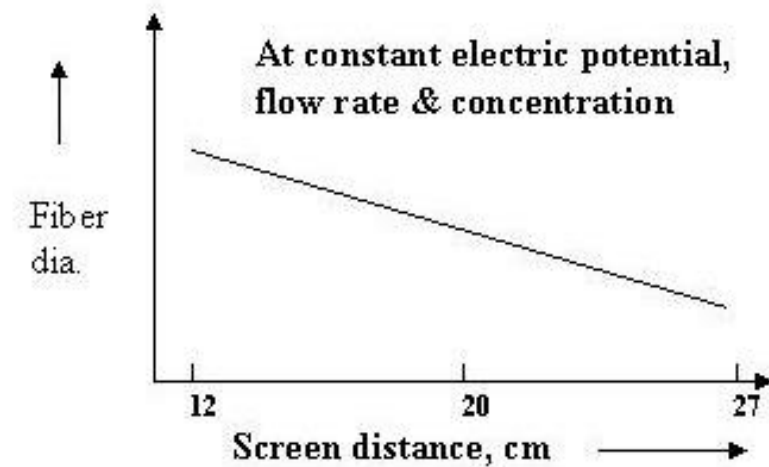
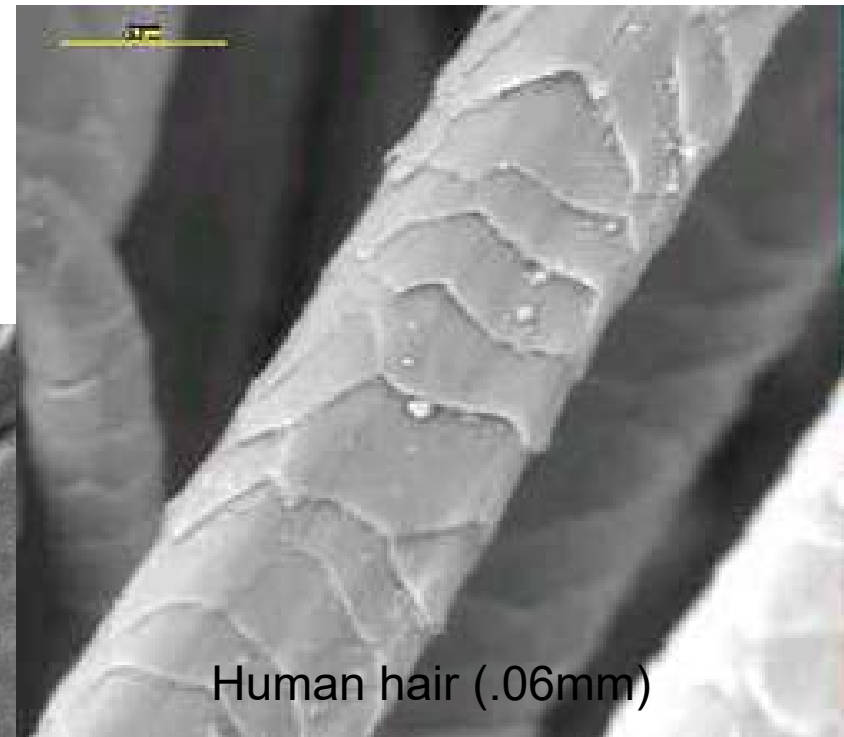
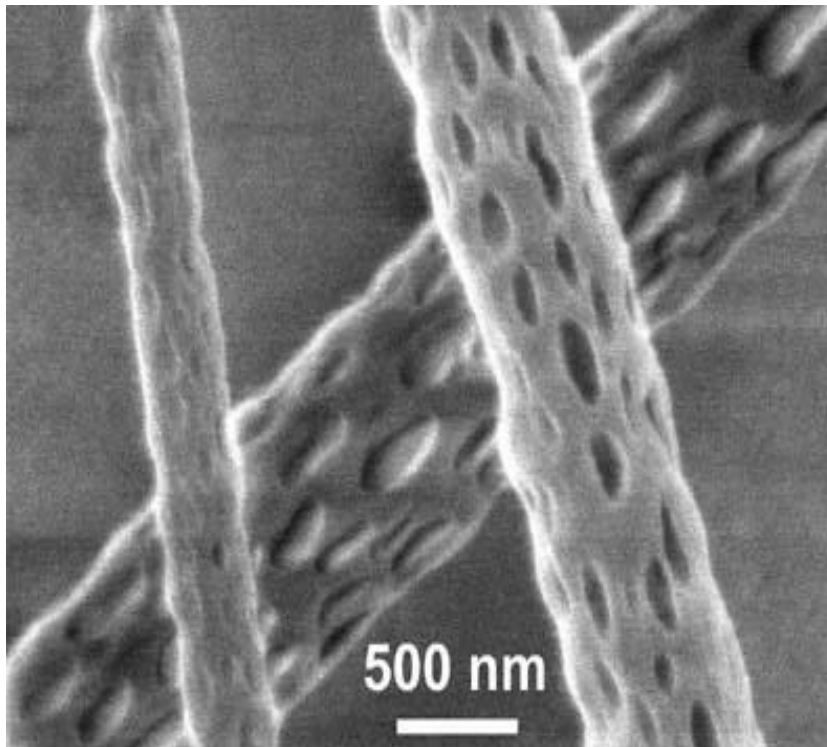
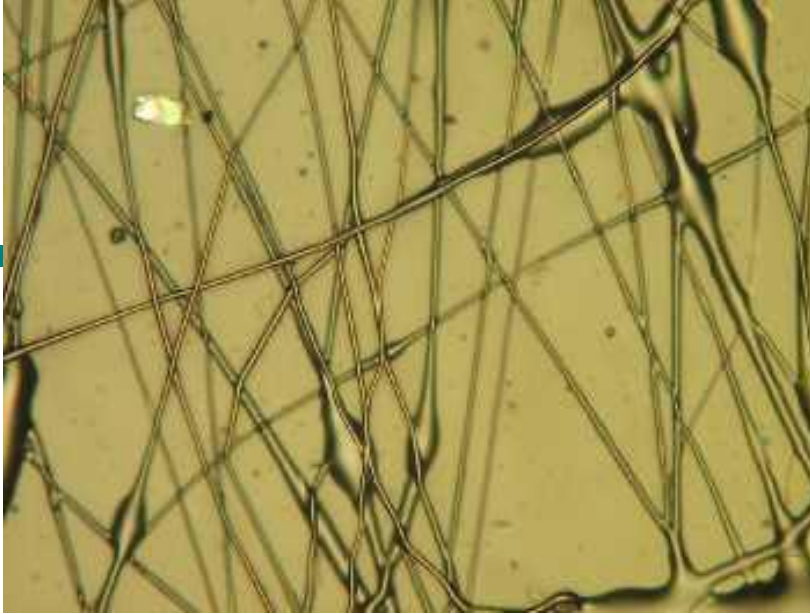


Figure 6. Effect of process parameters on fiber diameter, produced by Electrospinning

Electrospun polymers



Human hair (.06mm)

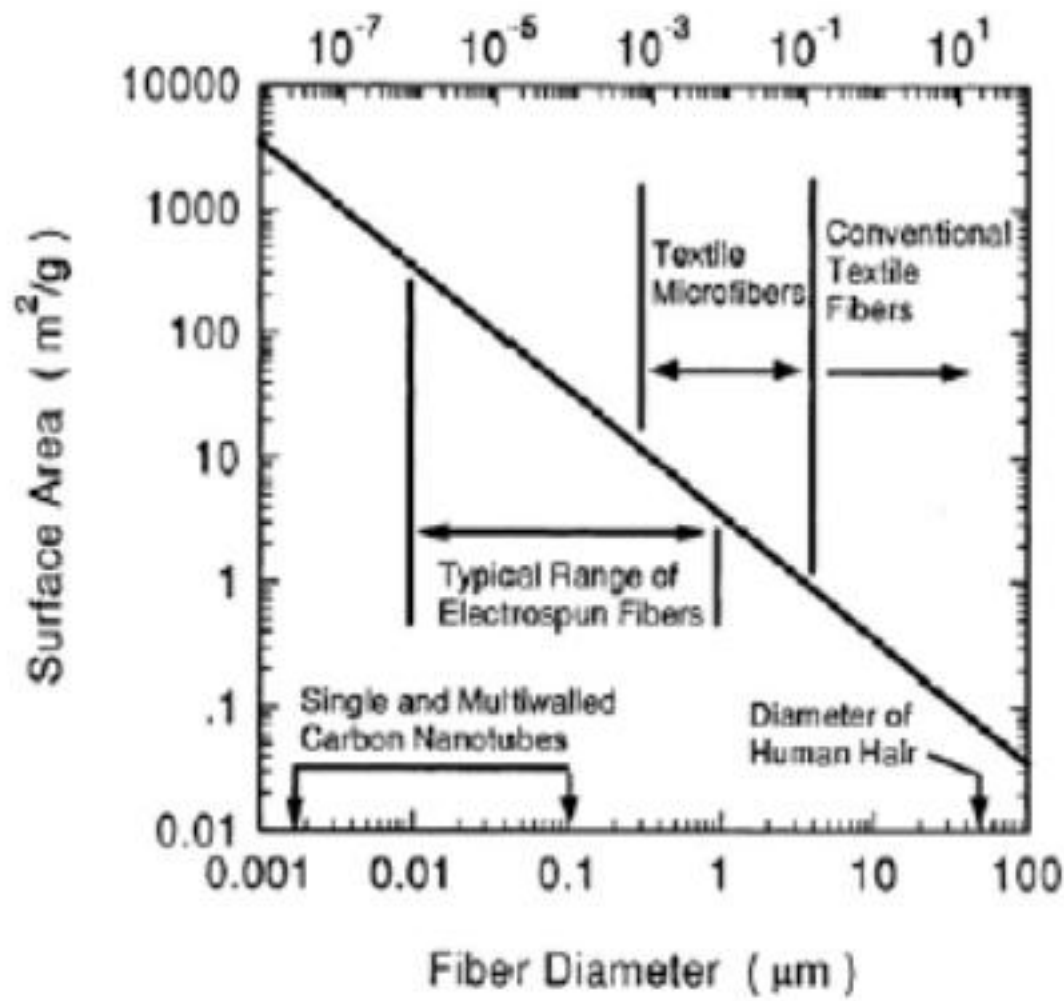
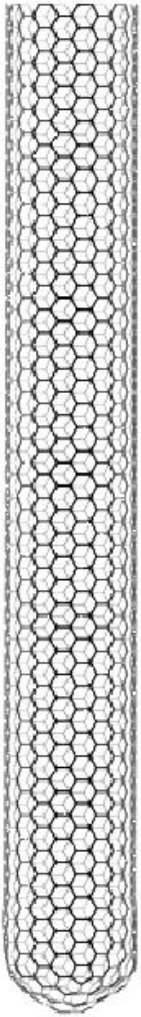


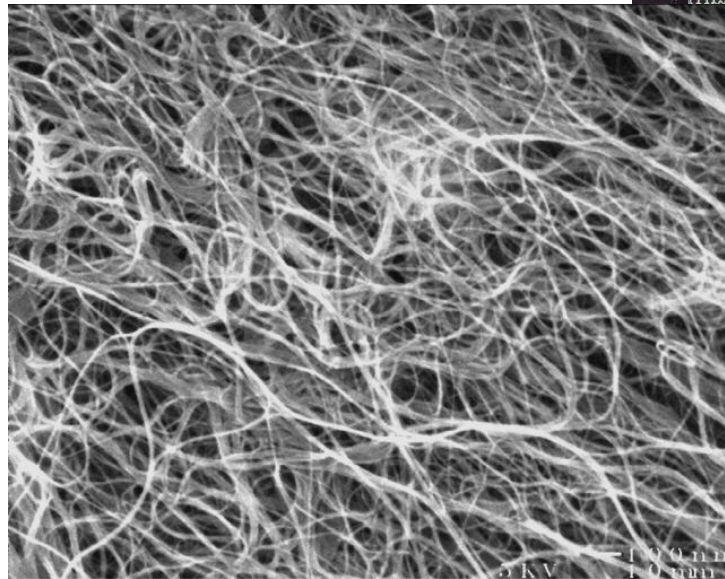
Figure 1: Effect of fiber diameter on surface area

Carbon Nanotube is Fiber



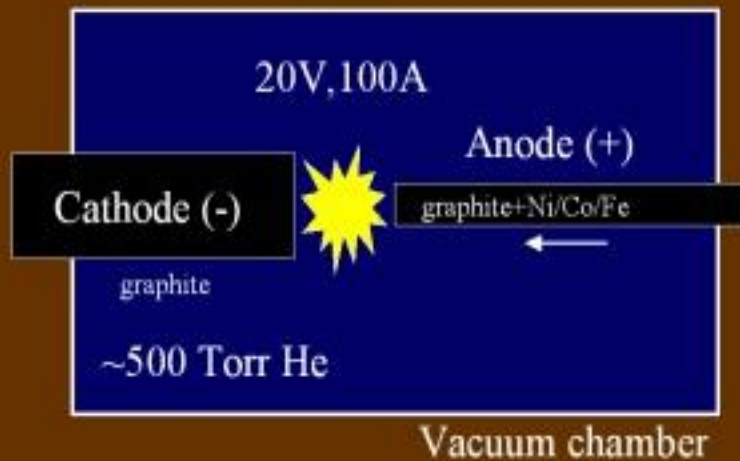
1000 X longer than diameter
Often uniaxial strength
Kevlar-strongest organic fiber

tensile strength 60GPa
Young's modulus 1TPa)

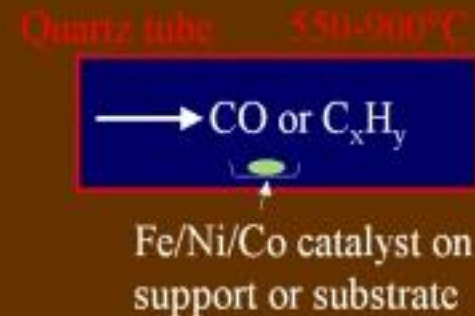


Making Carbon Nanotubes

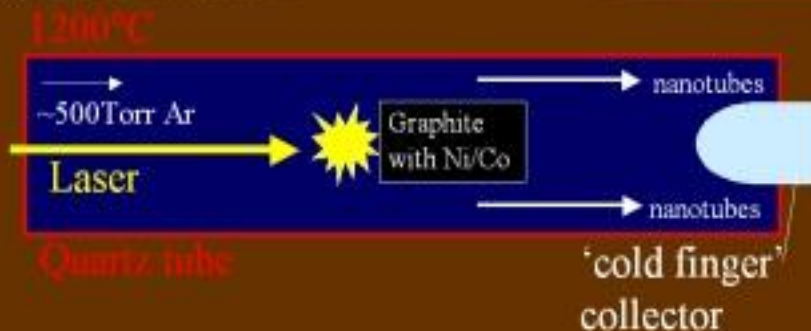
(a) Electric arc discharge



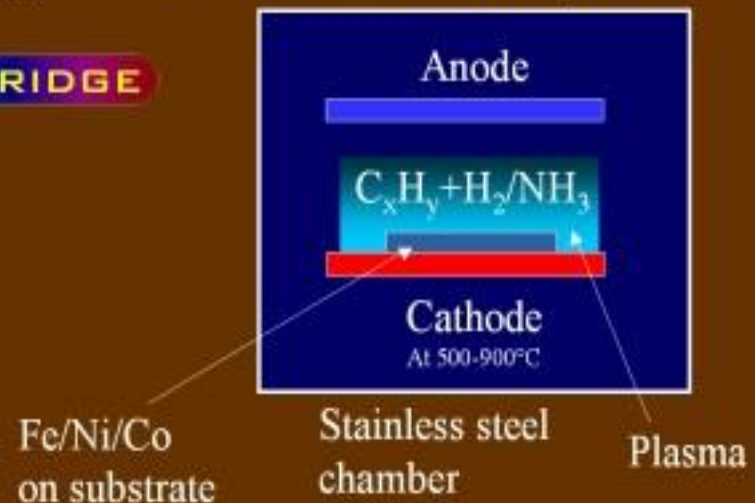
(c) Thermal chemical vapour deposition

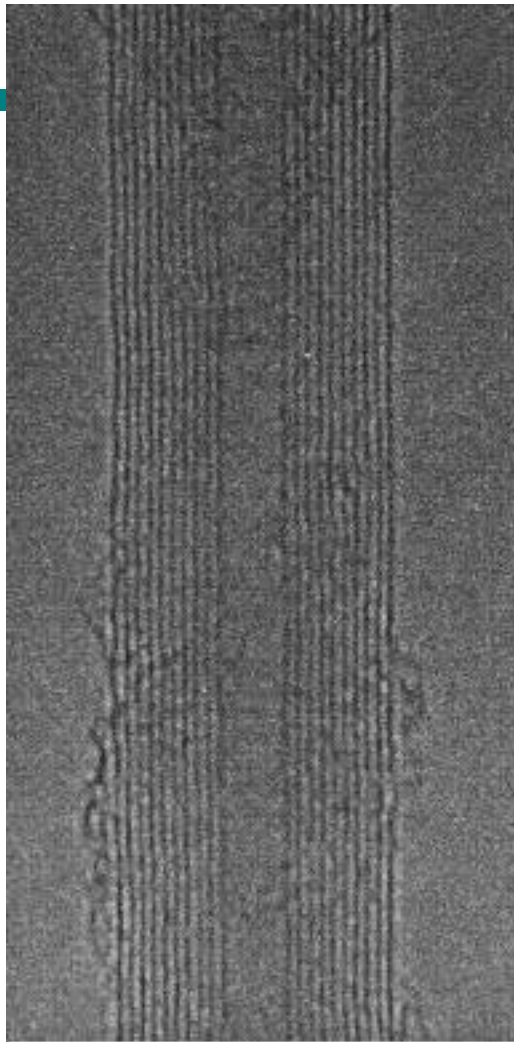


(b) Laser Ablation

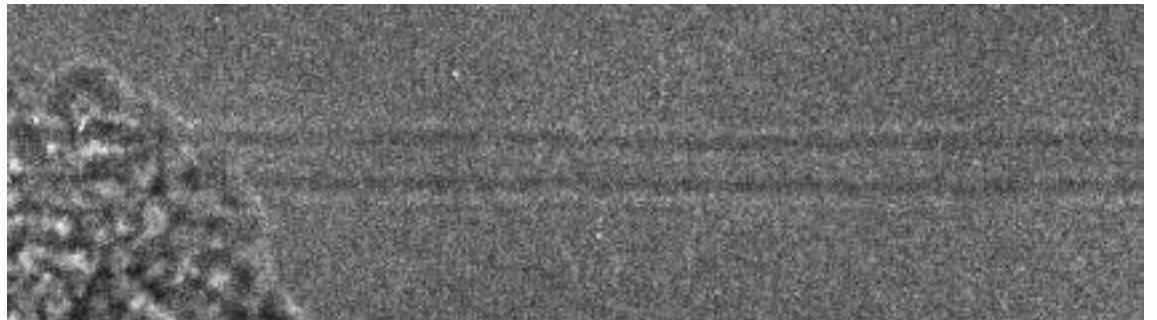


(d) Plasma enhanced chemical vapour deposition



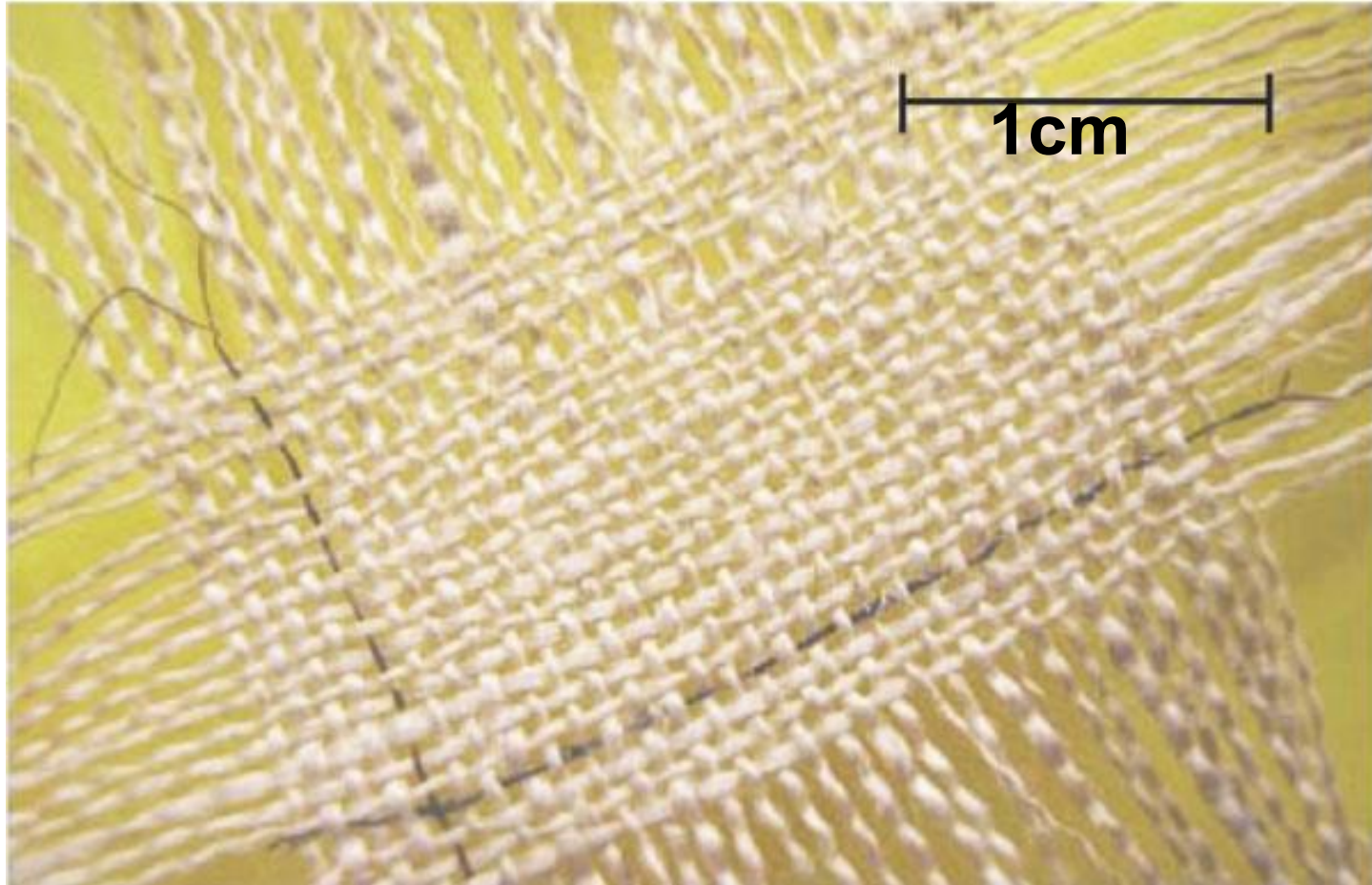


Above: TEM Image of a NWNT.
From the Zettl Research Group
(21)

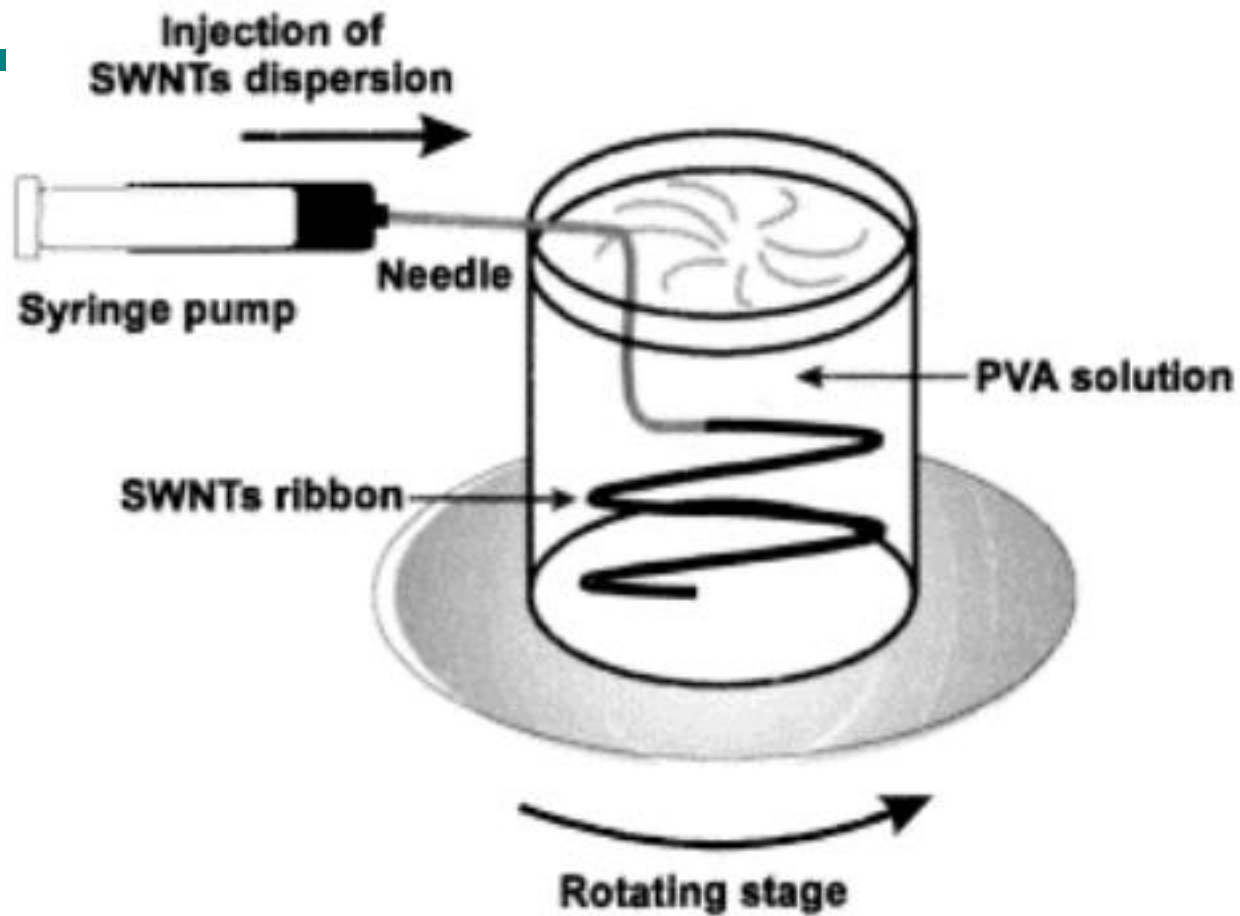


Above: TEM Image of a NWNT. From the Zettl Research Group (21)

Carbon Nanotube Fibers



Nature 423, 703 (12 June 2003); doi:10.1038/423703a



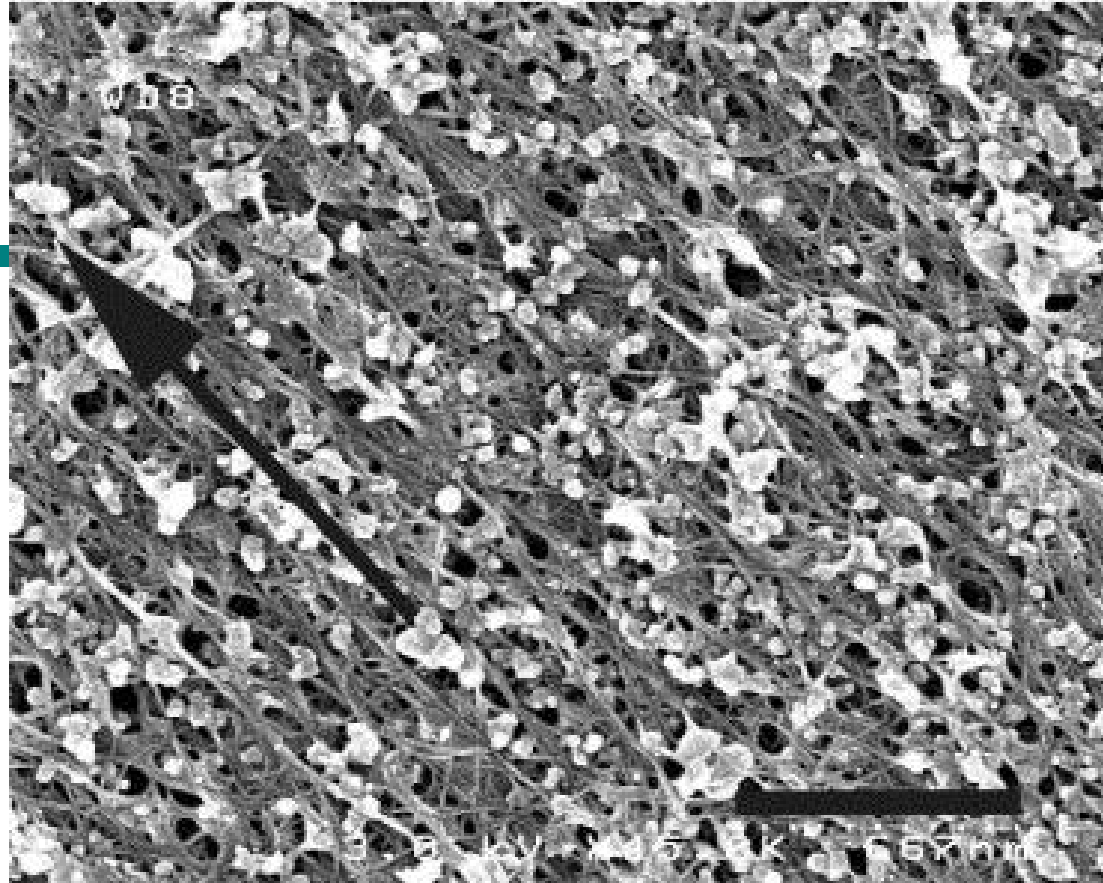
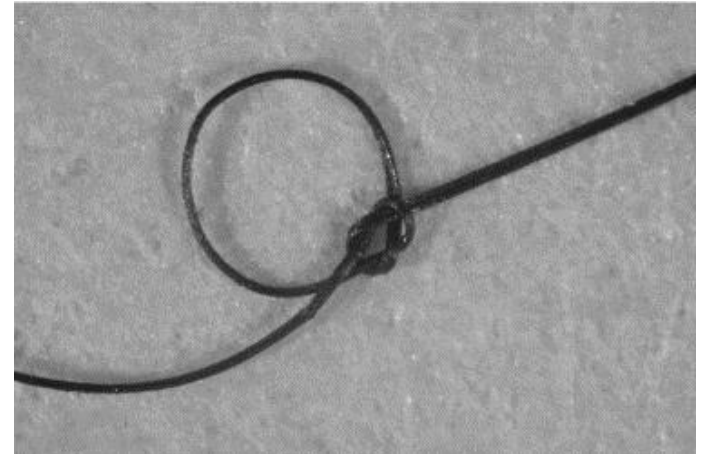
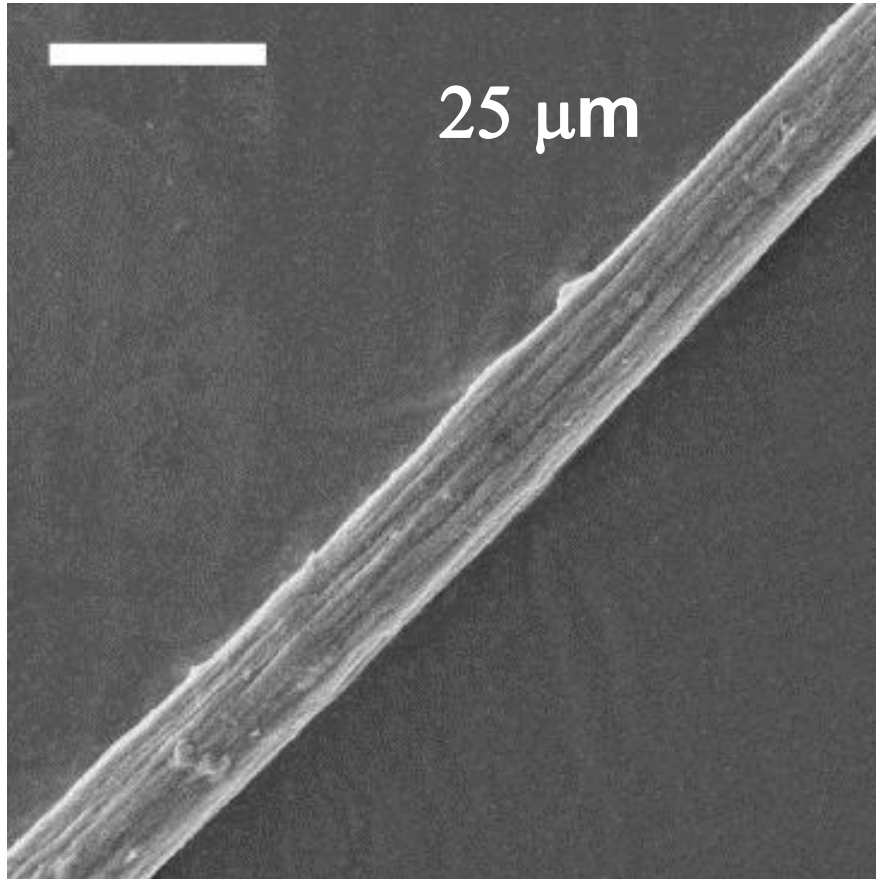


Fig. 4. Scanning electron micrograph of a dry ribbon deposited on a glass substrate. The black arrow indicates the main axis of the ribbons, which corresponds to the direction of the initial fluid velocity. Despite the presence of a significant amount of carbon spherical impurities, SWNTs bundles are preferentially oriented along the main axis. Scale BAR=667 nm

SWNT Fiber after drawing



Multifunctional Carbon Nanotube Yarns by Downsizing an Ancient Technology

Mei Zhang,¹ Ken R. Atkinson,² Ray H. Baughman^{1*}

By introducing twist during spinning of multiwalled carbon nanotubes from nanotube forests to make multi-ply, torque-stabilized yarns, we achieve yarn strengths greater than 460 megapascals. These yarns deform hysteretically over large strain ranges, reversible and irreversible. They are nearly as tough as fibers used in ropes and yarns, these nanotube yarns resist breaking by knotting. They also retain their strength after heating to 450°C for an hour or when immersed in water. They also have high electrical conductivity and are mechanically stable in air.

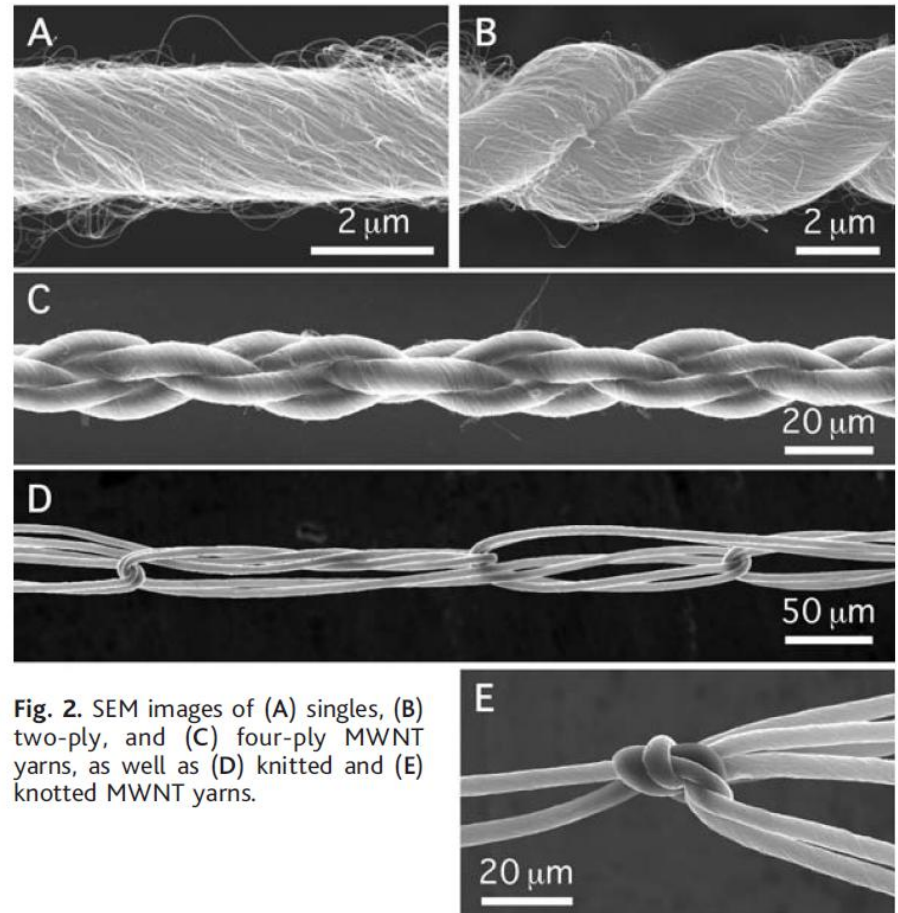
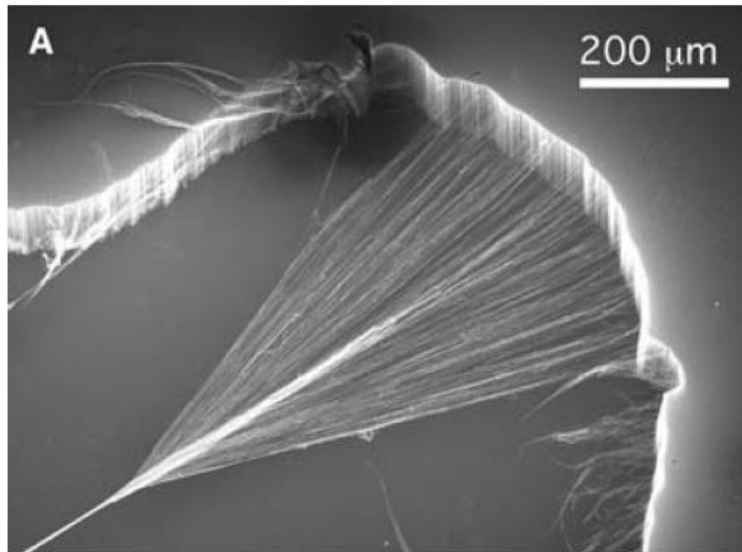


Fig. 2. SEM images of (A) singles, (B) two-ply, and (C) four-ply MWNT yarns, as well as (D) knitted and (E) knotted MWNT yarns.

HOMEWORK

- Introduce at least 3 polymer fibers and their applications
- Encourage you to give a short talk on what you find during next class.